

Critical Minerals, Geopolitics, and the Green Transition

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Abstract

The green energy transition will be powered by the mining and processing of lithium, nickel, and cobalt, which are critical for the production of advanced batteries. These minerals are concentrated geographically but traded globally, allowing key mining countries to exercise market power through policy intervention. Advanced batteries use multiple minerals in combination, and this joint use creates interdependence across mineral markets. We study the geopolitical implications of these forces and their consequences for green technology adoption worldwide. We quantify supply chain vulnerability, international policy spillovers, and the potential for mineral cartels.

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1 Introduction

The energy transition calls for the large-scale adoption of advanced battery technologies. This green future depends crucially on the production of lithium, nickel, and cobalt, which are essential inputs for advanced batteries. These critical minerals are traded in global markets, but their geological endowments are highly concentrated: the top three producing countries account for 70 to 85% of world output, far exceeding the concentration observed in fossil fuel markets. This concentration allows key mining countries to exercise market power by restricting upstream supply. At the same time, advanced batteries use multiple mineral inputs in combination, and this joint use creates interdependence across mineral markets. A policy that restricts one mineral affects not only the price and producers of that mineral, but also the prices and producers of other minerals used in the same batteries. We ask how market power and interdependence in upstream mineral markets shape the geopolitics of critical minerals and green technology adoption worldwide.

Geological endowments make upstream concentration an intrinsic feature of this setting. Mineral production is dominated by a small number of countries: Australia for lithium, Indonesia for nickel, and the Democratic Republic of the Congo (DRC) for cobalt. Production costs vary widely across mines due to persistent differences in ore grade, deposit types, and access to infrastructure. Battery technologies, in turn, vary in their mineral requirements. We focus on three main battery technologies: lithium iron phosphate (LFP), nickel manganese cobalt (NMC), and nickel cobalt aluminum (NCA). Each differs in its use of lithium, nickel, and cobalt as inputs.

This collection of battery technologies allows for both substitution across batteries and complementarity across minerals. Suppose Indonesia restricts nickel supply in order to raise nickel prices. A rise in nickel prices triggers two opposing forces for lithium demand. Substitution shifts battery demand toward LFP, which uses lithium but not nickel. Lithium demand rises. Complementarity operates within NMC and NCA, which use lithium and nickel together. As substitution lowers demand for these technologies, their use of both minerals declines. Lithium demand falls. The relative strengths of these forces determine the net effect on lithium prices. This net effect matters for who gains and loses from strategic mineral policy and how global green adoption responds. Whether substitution or complementarity dominates is an

empirical question.

To this end, we build a global dynamic equilibrium model that captures these elements. On the demand side, we focus on electric vehicle (EV) batteries, which account for 80% of annual installed battery capacity worldwide. We estimate an almost ideal demand system for EV manufacturers, using regional data on battery installations by technology. The data span the universe of EV models produced worldwide from 2021 to 2025. The demand system captures how manufacturers substitute across battery technologies in response to changes in battery prices. Battery technologies combine multiple mineral inputs according to recipes dictated by chemical properties, and we observe these recipes with engineering data. The recipes govern the strength of complementarity across minerals, which interacts with substitution across batteries to determine green adoption.

On the supply side, we estimate a dynamic model in which forward-looking mines choose production subject to capacity and finite reserves, using a mine-level panel dataset that covers roughly 90% of global lithium, nickel, and cobalt production. For each mine, we observe production, capacity, ore grade, ownership, and detailed cost data. We develop a general framework for dynamic supply estimation that leverages our cost data to address endogeneity bias. Intuitively, our average cost data provide an additional moment that we use to purge unobserved cost shocks, which otherwise lead to biased estimation. We account for existing policy wedges, planned mine entry, and calibrated reserve growth. The model captures the distribution of producer gains and losses in response to changes in mineral prices.

We apply our empirical framework to study supply disruptions and market power, both within and across mineral markets. First, we quantify supply chain vulnerability by removing the top producer from each mineral market. We observe the empirical signature of gross complementarity: removing a producer substantially raises its own mineral price, while also reducing other mineral prices. Consider the removal of Indonesian supply, which accounts for one-half of global nickel production. Nickel prices rise sharply, raising NMC and NCA battery prices, while lithium and cobalt prices fall, reducing LFP prices. These opposite effects contrast with the typical energy resource setting, in which competing countries produce substitute goods with prices that rise and fall together.

Second, we simulate unilateral mineral policy. Again consider Indonesia, which exercises its market power by reducing nickel supply and raising nickel prices. This policy action lowers lithium and cobalt prices through complementarity. In terms of policy spillovers, Indonesian interests align with those of other nickel producers, which are substitutes that benefit and free-ride as nickel prices rise. But Indonesian interests conflict with those of lithium and cobalt producers, which are complements that suffer as their mineral prices fall. In terms of green adoption, nickel-intensive NMC and NCA batteries become more expensive, while lithium-intensive LFP batteries become less expensive. Indonesian policy effectively taxes the former and subsidizes the latter, and consumers substitute accordingly. Total green adoption declines, but substitution softens the fall.

Third, we assess the impacts of mineral cartels. Single-mineral cartels coordinate among substitute producers within a given mineral, amplifying the effects of unilateral policy and unambiguously slowing the green transition. Multi-mineral cartels coordinate across complementary minerals with countervailing effects. On one hand, coordinating across minerals extends a multi-mineral cartel’s market power over a larger share of battery inputs. This force raises battery prices and reduces adoption. On the other hand, a unified cartel internalizes cross-mineral spillovers and thus sets less aggressive policy than separate competing cartels. The merger of complementary producers leads to the removal of double markups, reflecting the classic logic of Cournot complementarity ([Cournot 1838](#)). This force reduces battery prices and raises adoption. When inputs are complementary, consolidation can accelerate the green transition.

Fourth, we turn from country-level coordination through cartels to firm-level coordination through multinational ownership. Chinese multinationals hold significant mining stakes across all three mineral markets. Horizontal coordination across this portfolio internalizes complementarity, while vertical coordination internalizes the losses that higher mineral prices impose on China’s large downstream EV sector. Both temper China’s incentive to restrict upstream supply. We illustrate the geopolitical implications of these forces by simulating resource nationalization. In this counterfactual, the governments of leading mineral-producing countries expropriate mines from foreign multinationals and set policy independently. Nationalization severs cross-market ownership links, eliminating multinational coordination

across mineral markets while consolidating mines under domestic control within mineral markets. We find that nationalization amplifies market power and slows green adoption.

Our main contribution is to quantify market power and interdependence in natural resource markets. We connect most directly to studies of market power in individual resource markets ([Aguirregabiria and Luengo 2016](#), [Asker et al. 2019, 2024](#), [Kellogg 2024](#), [De Cannière 2025](#)), including work on oil markets and the Organization of the Petroleum Exporting Countries (OPEC).¹ We ask how market power will shape the next generation of energy resources. Our key departure from this literature is joint use: advanced batteries combine multiple minerals, creating interdependence across mineral markets with no direct analog in single-resource settings. This interdependence reflects both complementarity across minerals and substitution across batteries. We show how it changes the incidence of mineral market power and its consequences for green adoption.

We also contribute to a broad literature on trade and the environment, particularly as it relates to the environmental impacts of unilateral policy ([Rauscher 1997](#), [Copeland and Taylor 2003](#), [Kortum and Weisbach 2017](#), [Copeland et al. 2022](#), [Hsiao 2026](#), [Kortum and Weisbach 2026](#)). We show how upstream mineral policy propagates through global trade, distributing gains and losses across countries and shaping green adoption worldwide. Furthermore, our focus on market power and interdependence relates to a growing literature on geoeconomics ([Hirschman 1945](#), [Blackwill and Harris 2016](#), [Farrell and Newman 2019](#), [Becko et al. 2025](#), [Flynn et al. 2025](#), [Mohr and Trebesch 2025](#), [Clayton et al. 2026a,b](#)). We complement the largely macroeconomic and theoretical focus of this work with new microdata and quantitative evidence for the critical mineral markets that underpin the green transition.

Finally, we relate to work on the green transition ([Acemoglu et al. 2016](#), [Boer et al. 2024](#), [Butters et al. 2025](#), [Elliott 2026](#), [Gowrisankaran et al. 2026](#)), particularly recent work on the EV industry ([Springel 2021](#), [Barwick et al. 2025, 2026](#), [Kwon 2025](#), [Wang 2025](#), [Allcott et al. 2026](#), [Head et al. 2026](#), [Remmy 2026](#)). These studies largely focus on the downstream stages of the EV supply chain, including the impact of

¹ A broader set of papers examines the impacts of trade ([Farrokhi 2020](#), [Bornstein et al. 2023](#), [Abuin 2025](#)), innovation ([Popp 2002](#), [Hassler et al. 2021](#), [Alfaro et al. 2025](#)), and sanctions ([Moll et al. 2023](#), [Bachmann et al. 2024](#), [Johnson et al. 2026](#)).

subsidies on final consumers. We focus instead on mining and processing upstream, showing how upstream concentration affects downstream adoption. A small set of mineral-endowed countries will shape our progress toward a greener future. We study the geopolitics of that green transition.

2 Background and Data

We introduce our setting and data. We document resource concentration at the country level, large heterogeneity in production costs across mines, and rapid growth in battery adoption over the last decade.

2.1 Institutional details

Critical minerals

Critical minerals have become a strategic priority for governments worldwide. The US Energy Act of 2020 defines criticality broadly, citing “consequences for economic or national security” and vulnerability to disruption ([USGS 2026](#)), while Canada’s Critical Minerals Strategy emphasizes the “energy transition” and “green economy” ([NRC 2022](#)). We abstract from minerals that are critical for national security, including rare earth elements that are more closely tied to military applications. We focus instead on minerals that are critical for the energy transition, which requires advanced batteries for deploying renewable energy at scale. In turn, battery production depends on access to these key mineral inputs.

Table 1 shows that battery production is a leading driver of global demand for lithium, cobalt, and nickel – our primary minerals of interest. Within batteries, EV batteries dominate at 80% of global battery demand.² EV batteries alone account for 65% and 45% of global demand for lithium and cobalt, respectively, with non-EV batteries adding another 20% and 25%. EV batteries are a smaller but fast-growing 11% share of global demand for nickel, which is also a key input for stainless steel production. We exclude graphite because synthetic graphite is already a common alternative to natural graphite, and so supply is not constrained by concentrated

² EV batteries account for 80% of annual installed battery capacity. Non-EV battery energy storage systems (BESS) account for another 15%, and portable electronics for 5%.

Table 1: World mineral demand (2022–2024)

	Lithium	Cobalt	Nickel	Graphite	Copper	Manganese
EV batteries	65%	45%	11%	10%	4%	1%
Non-EV batteries	20%	25%	-	15%	-	-
Ceramics and glass	4%	-	-	-	-	-
Metals/alloys	-	19%	67%	-	-	90%
Wiring/infrastructure	-	-	-	-	92%	-
Refractories	-	-	-	40%	-	-
Other	11%	11%	22%	35%	4%	9%

Values for lithium, cobalt, and nickel are from [IEA \(2024\)](#). Values for copper, manganese, and graphite are from [USGS](#), [IRENA](#), [BHP Insights](#), and [Fastmarkets](#).

geographic endowments. We similarly exclude copper and manganese, which have much smaller shares of demand from battery technologies.

Supply chain

The path from mineral deposits to electric vehicles spans three broad stages: upstream mining and processing, midstream refining and battery manufacturing, and downstream EV manufacturing. Each stage features geographic concentration, although the dominant countries and firms differ across stages.

Upstream. Mineral resources generally take the form of hard-rock deposits, although South American lithium is typically found in brine deposits. Mines excavate hard rock and pump brine to the surface. We refer to this extraction as “mining.” Mines then engage in beneficiation, which involves the initial, on-site treatment of extracted resources. We refer to these activities as “processing.” For hard rock, mineral ore is separated from waste rock. For brines, the extracted liquid is concentrated through evaporation in open-air pools. The resulting mineral products are sold in global markets. For lithium mines, the product is spodumene or lithium carbonate, depending on deposit type. For cobalt mines, it is cobalt hydroxide. For nickel mines, it is typically nickel matte or mixed hydroxide precipitate. Upstream production is dominated by Australia for lithium, the DRC for cobalt, and Indonesia for nickel. Furthermore, foreign firms have a significant ownership presence across global mines, including in these countries.

Midstream. Refiners use mineral concentrates to produce battery-grade chemical products, including lithium hydroxide and carbonate, cobalt sulfate, and nickel sulfate. We distinguish this “refining” from the on-site processing that occurs upstream. Battery manufacturers use refined minerals to produce cathodes, which are combined with anodes, electrolytes, and separators to form battery cells. We focus on cathode mineral inputs because cathodes account for 45 to 67% of battery cell material costs (Knehr et al. 2024a).³ Cathode chemistry is also the primary source of differentiation across battery technologies. China plays a major role in both mineral refining and battery manufacturing, as Chinese firms account for 65 to 80% of global refined lithium, cobalt, and nickel production and roughly 70% of global battery production (IEA 2024, 2025).

Downstream. EV manufacturers use battery cells to produce vehicles for sale to final consumers. The EV manufacturing industry is global, with production concentrated in China, the United States, Europe, South Korea, and Japan. The largest EV producers in these markets are BYD, Tesla, Volkswagen, Hyundai, and Nissan, respectively. China is the largest EV market by volume, but adoption continues to rise across Europe and North America.

Battery technologies

Two battery technologies dominate: nickel manganese cobalt (NMC) and lithium iron phosphate (LFP). Both use lithium, which moves between the cathode and anode when charging and discharging. In NMC batteries, nickel provides energy density, and cobalt provides thermal stability. In LFP batteries, lithium iron phosphate plays both roles. These batteries are imperfect substitutes. NMC batteries have higher cost and higher energy density, which sustains a longer driving range. They are more common in the higher-end vehicles that prevail in Europe and North America. LFP batteries have lower cost and lower energy density, resulting in a shorter driving range. They are better suited for dense urban markets, which are common in China.

LFP adoption is rising globally. Between 2021 and 2023, the LFP share of the EV market grew from 52% to 67% in China, from 4% to 6% in Europe, and from 2% to

³ By comparison, anodes account for only 13 to 17% of battery cell material costs. Furthermore, most anodes are made from graphite, and most graphite is synthetic rather than mined. We treat graphite as a non-mineral intermediate input.

7% in the United States (IEA 2024). We also consider nickel cobalt aluminum (NCA) batteries, which have more limited adoption and slower growth over time. Because they contain more nickel than NMC batteries, NCA batteries have even higher cost and even longer range. Together, LFP, NMC, and NCA batteries account for 98% of annual EV battery installations (BMI 2025).

2.2 Data sources

We compile detailed data on upstream mining and processing, drawing on three proprietary sources: BMI (2025), GlobalData (2025), and S&P Global (2025). We link these sources to construct a global panel dataset at the mine level, covering lithium, cobalt, and nickel from 2010 to 2024. Appendix A details how we link records across datasets, standardize output measures, and validate the resulting panel. Our final sample includes 406 mines, which account for 94% of world production for lithium, 94% for cobalt, and 79% for nickel. We observe production quantities for both mined ore and processed minerals, as well as production capacity, ore grade, mine type, and mining method. Ownership information allows us to measure foreign ownership across mines. Cost data measure the average costs of production, which we leverage in our empirical modeling. These costs are broken down by category, including mining costs, processing costs, and royalties. We observe monthly mineral prices over the sample period.

We do not directly observe midstream refining or battery manufacturing activity. However, we do observe the chemistries that govern the use of minerals in battery manufacturing, which we refer to as battery recipes. We obtain these scientific recipes for each battery technology from Argonne National Laboratory (Knehr et al. 2024b). We also observe prices at both ends of the midstream layer: prices for processed minerals and prices for battery technologies. The gap between them reflects midstream costs and potential markups, which we recover jointly in estimation.

Finally, we observe downstream EV manufacturing, which represents demand for batteries. EV sales data from BMI (2025) record battery installations by technology, make, model, country, and quarter. For example, we observe the number of NMC batteries installed in all Tesla Model S units produced in the United States in the second quarter of 2022. For each battery technology, we observe monthly battery prices from 2021 to 2025.

2.3 Stylized facts

Mineral endowments are geographically concentrated

For both minerals and fossil fuels, unequal geological endowments lead to concentrated production. Mineral markets are especially concentrated. Figure 1 shows that the market share of the top three producing countries ranges from 40 to 50% for fossil fuels, but a substantially higher 70 to 85% for mineral commodities. This pattern also holds when concentration is measured by reserves instead of production. Concentration is particularly high for lithium and cobalt, which are more nascent markets, and it remains high for nickel, which is a more mature market.

Production costs are heterogeneous across mines

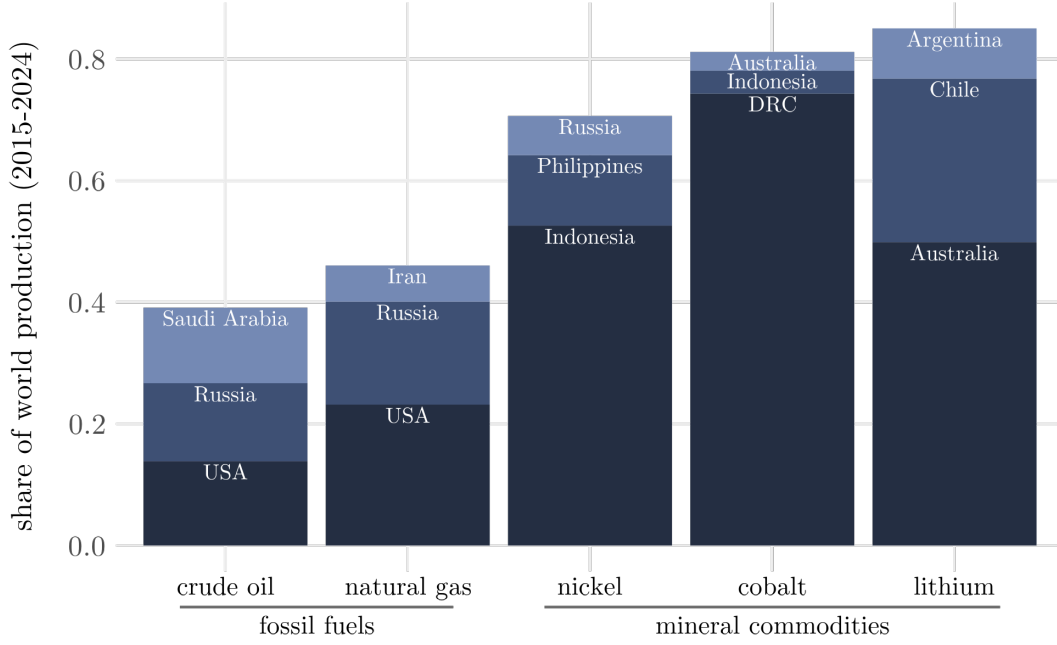
Our mine-level data reveal substantial heterogeneity in production costs across mines. Geological fundamentals drive this heterogeneity. Across minerals, deposits vary in their accessibility. Lithium in South America exists in underground brine deposits, which require capital-intensive pumping to the surface, while cobalt in the DRC can be mined artisanally because it is already accessible at surface level. Within minerals, deposits vary in their purity. Figure 2 illustrates this fact for cobalt. Average ore grade in the DRC is five times as high as ore grade elsewhere (left panel), and higher ore grade translates directly into lower production costs (right panel).

Battery recipes vary by technology

Battery technologies differ in their mineral inputs. Figure 3a shows battery recipes for LFP, NMC, and NCA batteries. These recipes capture interdependence across minerals: lithium, cobalt, and nickel are used jointly in producing NMC and NCA batteries, and this joint use creates complementarity. Figure 3b shows the growing adoption of LFP and NMC batteries over the last decade. LFP market share grew sharply from 15% to 50% between 2020 and 2024, as the expiration of key patents allowed for rapid global diffusion.⁴ By 2024, LFP had surpassed NMC.

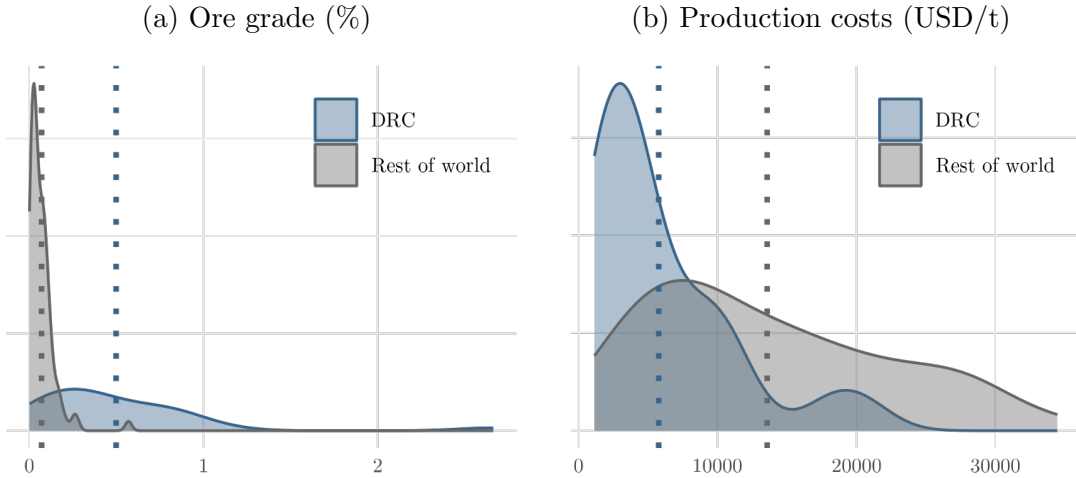
⁴ The core LFP patents were largely developed in North America. Follow-on advances enabled commercial-scale production in China, but licensing requirements impeded production elsewhere. The expiration of most patents by 2022 lowered barriers to global adoption.

Figure 1: Market concentration in world production (2015–2024)



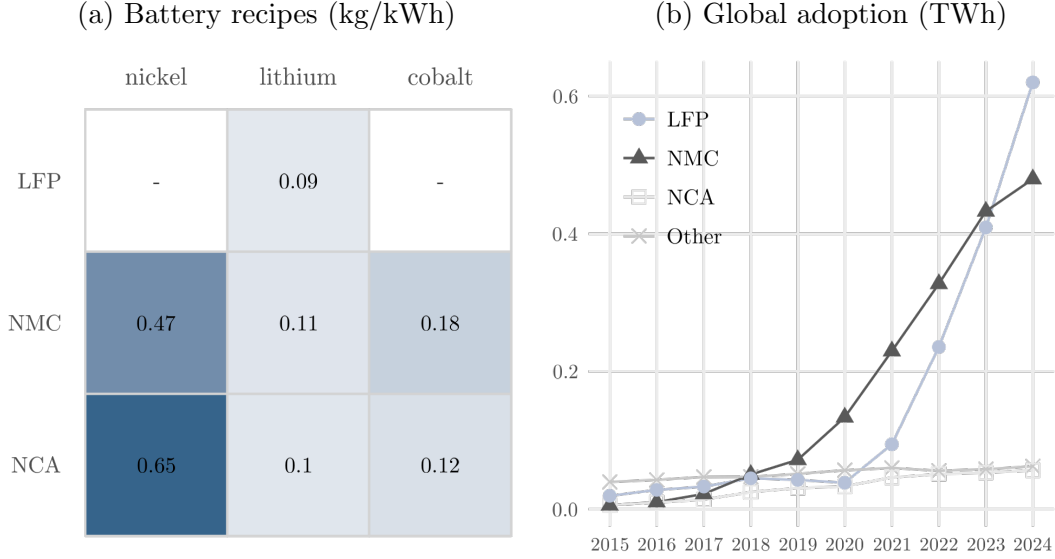
We aggregate production values to the country level from our mine-level data. Data on crude oil (including lease condensates) and natural gas come from [EIA \(2025\)](#).

Figure 2: Cobalt production across mines



On the left, we plot the density of average ore grades across mines. On the right, we plot the density of average production costs across mines. In both cases, we average over time for each mine.

Figure 3: Battery technologies



On the left, we plot the recipe matrix for selected battery technologies and minerals. NMC values average across all NMC varieties. Appendix A.2 shows the full recipe matrix. On the right, we plot newly installed capacity worldwide for each battery technology with data from BMI (2025).

3 Theory

We present a stylized model with two minerals, which serve as joint inputs for battery production. We characterize how this joint use creates interdependence across minerals and, in turn, shapes the consequences of mineral market power at the country level. We highlight implications for policy spillovers and green adoption.

Minerals and batteries

We focus on lithium and nickel. Consider two mining countries $m \in \{\ell, n\}$ and two battery technologies $j \in \{L, N\}$. Country ℓ produces only lithium, and country n produces only nickel. Battery production combines these minerals according to scientific recipes with a Leontief structure. We define r_j^m as the units of mineral m required to produce one unit of battery j , adopting superscripts for upstream minerals m and subscripts for downstream batteries j . Battery L uses only lithium with $r_L^\ell > 0$ and $r_L^n = 0$, while battery N uses both lithium and nickel with $r_N^\ell > 0$ and $r_N^n > 0$.

Recipe matrix $\mathbf{R} = [r_j^m]$ collects these terms.

$$\begin{array}{cc} & \begin{array}{cc} \text{Lithium } \ell & \text{Nickel } n \end{array} \\ \begin{array}{c} \text{Battery } L \\ \text{Battery } N \end{array} & \left[\begin{array}{cc} r_L^\ell & 0 \\ r_N^\ell & r_N^n \end{array} \right] \end{array}$$

Off-diagonal r_N^ℓ governs joint use, which has two dimensions. First, lithium is a common input in battery L and N production, as given by (r_L^ℓ, r_N^ℓ) . Second, lithium and nickel are complements in battery N production, as given by (r_N^ℓ, r_N^n) .

Supply and demand

Mining countries supply minerals $m \in \{\ell, n\}$. Country m supplies s^m units of its mineral as a function of world price p^m and mineral policy τ^m .

$$s^m(p^m - \tau^m), \quad \sigma^m = \frac{ds^m}{d(p^m - \tau^m)} > 0$$

We denominate the value of mineral policy in dollars. Restrictive policy τ^m reduces the quantity supplied with the goal of raising world prices.

Consumers demand batteries $j \in \{L, N\}$. Batteries are substitutes, and demand d_j is given by

$$d_j(p_L, p_N), \quad \delta_{jj} = \frac{\partial d_j}{\partial p_j} < 0, \quad \delta_{jj'} = \frac{\partial d_j}{\partial p_{j'}} > 0 \quad \text{for } j \neq j'.$$

When consumers can switch across batteries, own-price demand effects are negative and cross-price effects are positive. We maintain the standard regularity condition that demand Jacobian $\mathbf{D} = [\delta_{jj'}]$ is negative semidefinite, such that own-price effects dominate cross-price effects in magnitude. Furthermore, when battery prices rise, total battery demand d falls.

$$d(p_L, p_N) = d_L(p_L, p_N) + d_N(p_L, p_N), \quad \delta_j = \frac{\partial d}{\partial p_j} < 0$$

In equilibrium, mineral prices clear mineral markets. We have specified mineral

supply and battery demand, which yield mineral demand as follows.

$$d^\ell = d_L r_L^\ell + d_N r_N^\ell, \quad d^n = d_N r_N^n$$

Prices equalize supply and demand for each mineral.

$$s^m(p^m - \tau^m) = d^m(p^\ell, p^n) \quad \forall m \in \{\ell, n\} \quad (1)$$

If battery manufacturing is perfectly competitive, then mineral prices and battery recipes pin down battery prices.

$$p_L = p^\ell r_L^\ell, \quad p_N = p^\ell r_N^\ell + p^n r_N^n \quad (2)$$

Substitutes and complements

Mineral markets are linked on the demand side by substitution across batteries and complementarity across minerals. The cross-price effect of nickel prices on lithium demand is

$$\frac{\partial d^\ell}{\partial p^n} = \delta_{LN} r_L^\ell r_N^n + \delta_{NN} r_N^\ell r_N^n. \quad (3)$$

The net effect depends on the force that dominates. On one hand, $\delta_{LN} r_L^\ell r_N^n > 0$ captures substitution. Higher nickel prices raise the price of battery N , and so demand shifts toward battery L , which becomes cheaper in relative terms. Lithium demand rises as L expands and uses more lithium. On the other hand, $\delta_{NN} r_N^\ell r_N^n < 0$ captures complementarity. Higher nickel prices reduce demand for battery N , which also uses lithium. Lithium demand falls as N shrinks and uses less lithium. The cross-price effect of lithium prices on nickel demand is symmetric: $\frac{\partial d^n}{\partial p^\ell} = \delta_{NL} r_L^\ell r_N^n + \delta_{NN} r_N^\ell r_N^n$.

Definition 1 (Gross substitutes). Lithium and nickel are gross substitutes when substitution dominates, such that $\frac{\partial d^\ell}{\partial p^n} > 0$.

Definition 2 (Gross complements). Lithium and nickel are gross complements when complementarity dominates, such that $\frac{\partial d^\ell}{\partial p^n} < 0$.

Policy spillovers

We study the distributional impacts of policy action τ^n , which country n implements unilaterally. Differentiating equilibrium condition 1 with respect to τ^n , we

obtain the own- and cross-price effects of restrictive nickel policy.

$$\frac{dp^n}{d\tau^n} = \frac{\sigma^n \left(\sigma^\ell - \frac{\partial d^\ell}{\partial p^\ell} \right)}{\left(\sigma^\ell - \frac{\partial d^\ell}{\partial p^\ell} \right) \left(\sigma^n - \frac{\partial d^n}{\partial p^n} \right) - \frac{\partial d^\ell}{\partial p^n} \frac{\partial d^n}{\partial p^\ell}}, \quad \frac{dp^\ell}{d\tau^n} = \left(\frac{\frac{\partial d^\ell}{\partial p^n}}{\sigma^\ell - \frac{\partial d^\ell}{\partial p^\ell}} \right) \frac{dp^n}{d\tau^n}$$

Proposition 1 (Mineral effects). Restrictive nickel policy raises nickel prices. It raises lithium prices if and only if lithium and nickel are gross substitutes, and it reduces lithium prices if and only if lithium and nickel are gross complements.

Proof. For nickel prices, we show that $\frac{dp^n}{d\tau^n} > 0$. The denominator is positive: it is the determinant of market-clearing matrix $\mathbf{M} = \text{diag}(\sigma^\ell, \sigma^n) - \mathbf{R}'\mathbf{D}\mathbf{R}$ for recipe matrix $\mathbf{R} = [r_j^m]$, and negative semidefinite $\mathbf{D}$ implies negative semidefinite $\mathbf{R}'\mathbf{D}\mathbf{R}$ and positive definite $\mathbf{M}$, such that $\det \mathbf{M} > 0$. The numerator is also positive: negative semidefinite $\mathbf{D}$ implies $\frac{\partial d^\ell}{\partial p^\ell} = (\mathbf{r}^\ell)' \mathbf{D} \mathbf{r}^\ell \leq 0$ for $\mathbf{r}^\ell = (r_L^\ell, r_N^\ell)'$, such that $\sigma^\ell - \frac{\partial d^\ell}{\partial p^\ell} > 0$. For lithium prices, given $\frac{dp^n}{d\tau^n} > 0$ and $\sigma^\ell - \frac{\partial d^\ell}{\partial p^\ell} > 0$ above, the sign of $\frac{dp^\ell}{d\tau^n}$ is determined by the sign of $\frac{\partial d^\ell}{\partial p^n}$. Definitions 1 and 2 then give the result. $\square$

These policy effects have geopolitical implications. Country n exercises market power through τ^n , restricting nickel supply and raising nickel prices to increase its producer surplus. This action spills over to country ℓ through lithium prices. If lithium and nickel are gross substitutes, then country ℓ free-rides: it gains producer surplus as lithium prices rise. The countries' interests are aligned. By contrast, if lithium and nickel are gross complements, then country ℓ suffers: it loses producer surplus as lithium prices fall. The countries' interests are in conflict. Lithium policy has symmetric effects.

Green adoption

We turn to aggregate implications for the green transition. Batteries L and N are equally effective at reducing carbon emissions, and so the green transition depends only on total adoption $d = d_L + d_N$. Differentiating with respect to τ^m for a given mineral m , we obtain the effect of restrictive mineral policy.

$$\frac{dd}{d\tau^m} = \delta_L \frac{dp_L}{d\tau^m} + \delta_N \frac{dp_N}{d\tau^m}, \quad m \in \{\ell, n\}$$

Nickel and lithium policies do not have symmetric effects.

Proposition 2 (Battery effects of nickel policy). Without joint use ($r_N^\ell = 0$), restrictive nickel policy raises both battery prices and reduces total green adoption. With joint use ($r_N^\ell > 0$), it subsidizes battery L if lithium and nickel are gross complements, offsetting the reduction in total green adoption.

Proof. Without joint use, $r_N^\ell = 0$ implies that $\frac{\partial d^\ell}{\partial p^n} = \delta_{LN} r_L^\ell r_N^n > 0$, and so $\frac{dp^\ell}{d\tau^n}, \frac{dp^n}{d\tau^n} > 0$ by proposition 1. It follows that $\frac{dp_L}{d\tau^n}, \frac{dp_N}{d\tau^n} > 0$ by equation 2. Thus, $\frac{dd}{d\tau^n} = \delta_L \frac{dp_L}{d\tau^n} + \delta_N \frac{dp_N}{d\tau^n} < 0$ for $\delta_L, \delta_N < 0$. With joint use and gross complements, we have $\frac{dp^\ell}{d\tau^n} < 0$ by proposition 1 and thus $\frac{dp_L}{d\tau^n} = r_L^\ell \frac{dp^\ell}{d\tau^n} < 0$ by equation 2. This battery L subsidy is a pro-adoption force. $\square$

Proposition 3 (Battery effects of lithium policy). Without joint use ($r_N^\ell = 0$), restrictive lithium policy raises both battery prices and reduces total green adoption. With joint use ($r_N^\ell > 0$), it further taxes battery N and deepens the reduction in total green adoption.

Proof. Without joint use, $r_N^\ell = 0$ implies that $\frac{\partial d^\ell}{\partial p^n} = \delta_{LN} r_L^\ell r_N^n > 0$, and so $\frac{dp^\ell}{d\tau^\ell}, \frac{dp^n}{d\tau^\ell} > 0$ by proposition 1 and symmetry. Thus, $\frac{dp_L}{d\tau^\ell}, \frac{dp_N}{d\tau^\ell} > 0$ by equation 2, and $\frac{dd}{d\tau^\ell} < 0$ for $\delta_L, \delta_N < 0$. With joint use, we have $\frac{dp^\ell}{d\tau^\ell} > 0$ by proposition 1 and symmetry, and $\frac{dp_N}{d\tau^\ell} = r_N^\ell \frac{dp^\ell}{d\tau^\ell} + r_N^n \frac{dp^n}{d\tau^\ell}$ gains a positive term $r_N^\ell \frac{dp^\ell}{d\tau^\ell} > 0$ by equation 2. This additional battery N tax is an anti-adoption force. $\square$

Nickel and lithium policies diverge because of joint use. Without joint use, lithium and nickel are symmetric. Restricting either taxes both batteries, and green adoption falls. Joint use introduces asymmetry, and its effects depend on which mineral is restricted. Nickel and lithium are complements in battery N , and so restrictive nickel policy subsidizes battery L . Joint use helps adoption by encouraging substitution away from nickel. By contrast, lithium is a common input for batteries L and N , and so restrictive lithium policy taxes both batteries. Joint use hurts adoption by preventing substitution away from lithium. Our empirical analysis will quantify these opposing forces.

4 Demand

We model EV manufacturers' demand for battery technologies with an almost ideal demand system (Deaton and Muellbauer 1980). We estimate the model with regional panel data on battery technology use.

4.1 Model

In each region k and period t , EV manufacturers choose among battery technologies $j \in \mathcal{J}$. They do so by allocating total battery expenditures X_{kt} across batteries j , taking world prices p_{jt} as given.⁵ We specify their demand in product space with expenditure shares w_{jkt} given by

$$w_{jkt} = \alpha_{jt} + \beta_j \log \left(\frac{X_{kt}}{P_t} \right) + \sum_{j' \in \mathcal{J}} \gamma_{jj'} \log p_{j't} + \varepsilon_{jkt}, \quad (4)$$

where battery price index P_t is of translog form

$$\log P_t = \sum_{j \in \mathcal{J}} \alpha_{jt} \log p_{jt} + \frac{1}{2} \sum_{j, j' \in \mathcal{J}} \gamma_{jj'} \log p_{jt} \log p_{j't}. \quad (5)$$

Quality α_{jt} captures unobservables that vary across products and over time, but are common across regions. Income effects β_j allow expenditure shares to respond heterogeneously to changes in total battery expenditures. Price index P_t deflates total battery expenditures, and semi-elasticities $\gamma_{jj'}$ govern own- and cross-price elasticities. We allow for unobserved regional demand shocks ε_{jkt} .

We compute quantities demanded as follows. For battery prices $\mathbf{p}_t = \{p_{jt}\}_{j \in \mathcal{J}}$, the quantity demanded d_{jkt} of battery j in region k and period t is

$$d_{jkt}(\mathbf{p}_t) = \frac{X_{kt} w_{jkt}(\mathbf{p}_t)}{p_{jt}}$$

⁵ Total battery expenditures X_{kt} summarize EV demand, capturing the aggregate choice between EVs and non-EVs and the influence of consumer preferences, fossil fuel prices, and government subsidies. We condition on these nominal expenditures and characterize their allocation across battery technologies. Appendix B describes how real expenditures and battery quantities respond to battery prices in our counterfactuals.

by definition of expenditure share w_{jkt} . World demand aggregates across regions.

$$d_{jt}(\mathbf{p}_t) = \sum_{k \in \mathcal{K}} d_{jkt}(\mathbf{p}_t) \quad (6)$$

We specify demand directly over battery technologies, whose quantities map to mineral demand through the battery recipes.⁶ This approach captures the aggregate reallocation across batteries that is relevant for clearing mineral markets. It integrates over manufacturers’ battery choices within vehicle models, the reallocation of sales across existing models, and the extensive margin of model entry and exit. Separating these margins requires additional modeling structure, and our counterfactuals depend only on their combined effect.

4.2 Estimation

We estimate the model with iterated linear least squares (ILLS), as in [Blundell and Robin \(1999\)](#). The first estimation challenge is nonlinearity. Because of translog price index P_t , parameters α_{jt} and $\gamma_{jj'}$ enter nonlinearly in expenditure shares w_{jkt} . However, expenditure shares are linear in parameters when we condition on a fixed value of P_t . We exploit this linearity for estimation. First, we initialize estimation by computing P_t from data with the Stone price index, which is the expenditure-weighted average of prices p_{jt} . Second, we estimate expenditure share equation 4 by linear regression. Third, we use the resulting parameter estimates to recompute P_t with equation 5. Fourth, we iterate until convergence. Throughout, we impose the standard restrictions of adding-up, homogeneity, and symmetry. Appendix B provides additional details.

The second estimation challenge is price endogeneity: battery prices p_{jt} are correlated with demand shocks ε_{jkt} . We address this concern by instrumenting with cost shifters in the form of mineral prices. Relevance holds because minerals are inputs for batteries. Mineral prices shift battery costs and thus battery prices. Exclusion requires us to distinguish between two sets of minerals, $\mathcal{M}$ and $\mathcal{M}'$. The first set $\mathcal{M}$ consists of lithium, cobalt, and nickel, which are the main minerals of focus in

⁶ A common alternative is to model demand in characteristic space, as in [Berry et al. \(1995\)](#), particularly when the number of products is large. This alternative requires specifying the characteristics that govern substitution across products. We instead model demand in product space.

this paper. EV batteries are a major part of global demand for these minerals, and so their prices are likely correlated with battery demand shocks. The second set $\mathcal{M}'$ consists of aluminum, phosphoric acid, iron, and manganese. These minerals are inputs in EV battery production, but EV battery production is less than 3% of global demand for these minerals. Their prices are therefore plausibly uncorrelated with battery demand shocks. Appendix B shows that our estimates are robust to the set of minerals included in $\mathcal{M}'$.

We therefore instrument for battery prices p_{jt} with mineral prices $\{p^{mt}\}_{m \in \mathcal{M}'}$ from the second set $\mathcal{M}'$. We do so by constructing predicted prices z_{jt} , which weight input minerals according to their use in battery recipes r_j^m .

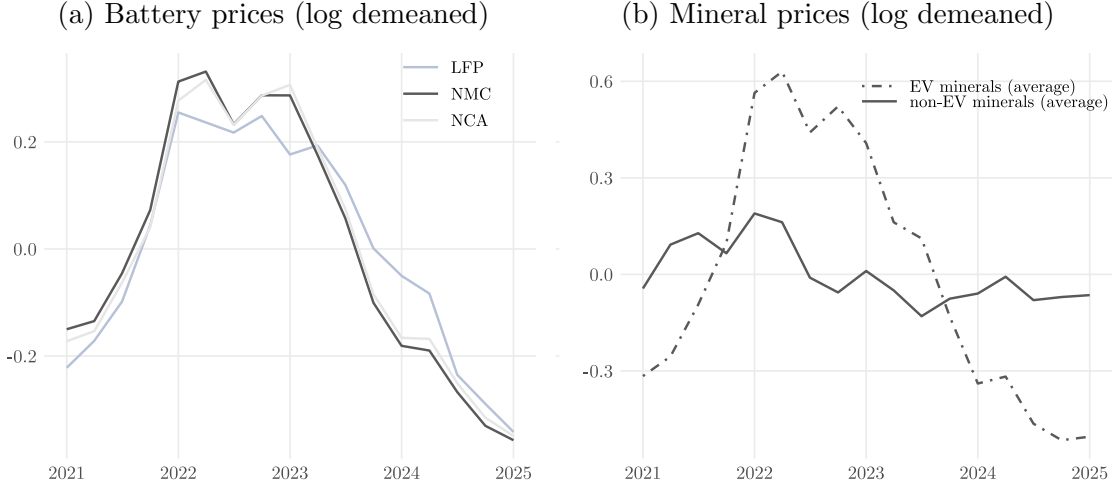
$$z_{jt} = \sum_{m \in \mathcal{M}'} r_j^m p^{mt} \quad (7)$$

Figure 4 shows that mineral prices in the first set $\mathcal{M}$ move closely with battery prices, in contrast to those in the second set $\mathcal{M}'$. Our instrument leverages the second, more orthogonal source of variation. Figure 4 also shows that battery prices move closely with each other. Our instrument disentangles these battery prices with battery-specific recipes r_j^m , by which variation in mineral prices generates differential variation in battery prices.

Our estimation sample is a quarterly panel from 2021 Q4 to 2025 Q1 with four regions of EV manufacturers: China, North America, Europe, and Asia ex-China. Appendix B assesses sensitivity to the quarterly frequency with two-quarter aggregation and rolling-average prices, yielding broadly similar results. Battery prices are in units of USD per MWh of capacity. The choice set includes two battery technologies that account for 98% of global EV battery installations: battery L is LFP, and battery N aggregates over NMC and NCA.⁷ This grouping aligns to first order with the mineral endowments that drive our geopolitical impacts of interest. Mineral policy in nickel- and cobalt-endowed countries must consider substitution away from NMC and NCA batteries, which use nickel and cobalt, toward LFP batteries, which do not.

⁷ We omit the remaining 2%, which consists of EV battery technologies with limited commercial deployment (e.g., sodium-ion). We combine NMC and NCA because their recipes and prices are particularly highly correlated, as seen in Figures 3 and 4. The aggregate expenditure share sums over NMC and NCA. The aggregate recipe and price are expenditure-weighted averages. Because NCA accounts for only a small share of installations, it receives little weight in these aggregates.

Figure 4: Battery and mineral prices



Mineral prices are from [S&P Global \(2025\)](#), and battery prices are from [BMI \(2025\)](#). All prices are averaged at the quarterly level and log demeaned. Panel (b) shows the average prices of EV minerals $\mathcal{M}$ and non-EV minerals $\mathcal{M}'$.

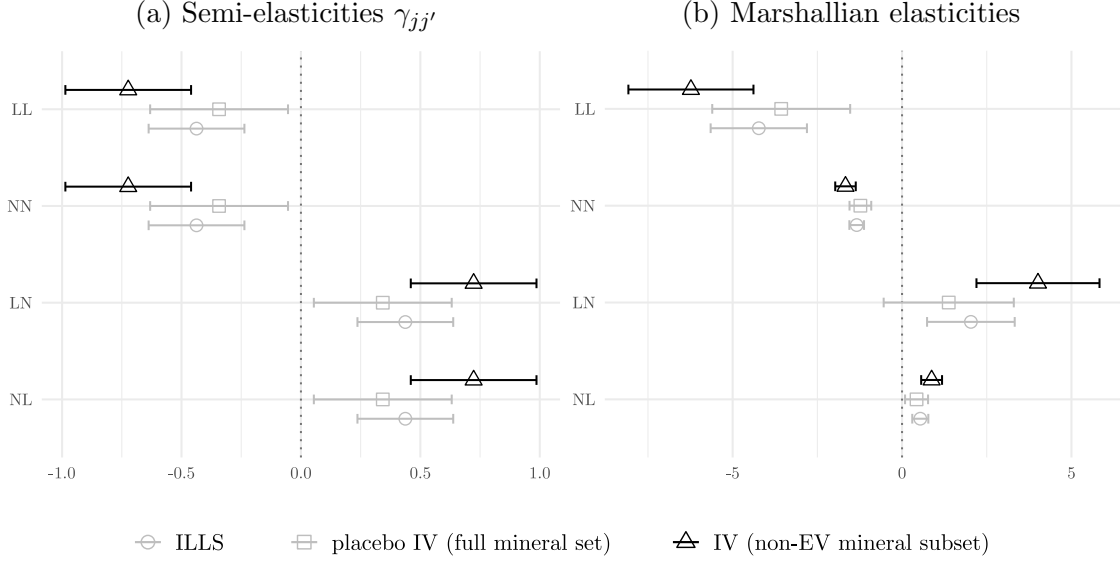
4.3 Estimates

Figure 5 reports our estimated demand parameters and elasticities. We begin with the estimates from our baseline IV specification, which the figure displays in black. Panel (a) shows that the own-price semi-elasticities γ_{jj} are negative, while the cross-price semi-elasticities $\gamma_{jj'}$ are positive. These parameter estimates determine the Marshallian elasticities in panel (b), with similarly negative own-price elasticities (LL and NN) and positive cross-price elasticities (LN and NL). We find that demand is elastic for both L and N , with elasticities that exceed one in magnitude, and that demand is particularly elastic for L . This result is consistent with the use of LFP batteries in lower-end vehicles, which have shorter driving ranges and serve more price-sensitive consumers.⁸

We compare our baseline IV estimates to two additional benchmarks in Figure 5: estimation without instruments and a placebo IV that uses the full set of minerals $\mathcal{M} \cup \mathcal{M}'$. Estimation without instruments suffers from price endogeneity, as demand

⁸ Appendix B reports estimates for parameters α_{jt} and β_j . We estimate lower quality α_{jt} for battery L , rationalizing lower shares despite lower prices in the data, and larger income effects β_j for L , reflecting that rising aggregate battery expenditure coincides with the rapid growth of LFP adoption during our sample period.

Figure 5: Demand estimates



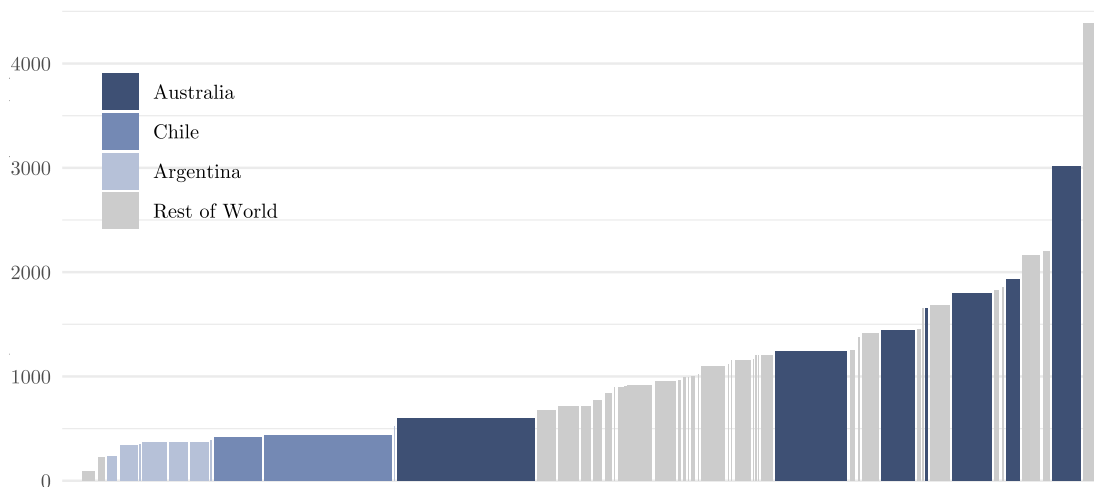
Vertical axes label the subindices of each parameter. We compute Marshallian elasticities as $\frac{\partial \log d_j}{\partial \log p_{j'}} = \frac{1}{\bar{w}_j}(\gamma_{jj'} - \beta_j \bar{w}_{j'}) - \mathbb{1}[j = j']$, where $\bar{w}_j$ is the worldwide expenditure share averaged over time for battery j . We cluster standard errors by quarter, allowing for serial correlation with a lag of one quarter. We compute Marshallian elasticity standard errors using the delta method.

shocks and battery prices are positively correlated. Panel (b) shows that the resulting estimates are biased toward zero. By comparison, our baseline IV estimates are roughly twice as large in magnitude. The placebo IV adds the first set of minerals $\mathcal{M}$ to the instruments. EV demand drives global demand for these minerals, and so their prices remain correlated with battery demand shocks ε_{jkt} . Instruments based on $\mathcal{M}$ thus fail the exclusion restriction, and panel (b) confirms that the placebo estimates remain similar to the estimates without instruments. Our baseline IV corrects this simultaneity bias by excluding $\mathcal{M}$ from the instrument set.

5 Supply

We model the supply of minerals with a dynamic model of extraction, and we estimate the model with mine-level panel data on production quantities and costs. The model delivers country-level supply curves that characterize the distribution of producer rents across mineral-endowed countries.

Figure 6: Lithium production costs across mines (USD/t)



Mine-level cost data for 2024 come from [BMI \(2025\)](#). Each bar is a mine. Bar heights indicate realized average costs in USD per tonne of lithium carbonate equivalent. Bar widths are proportional to production capacities.

5.1 Cost data

We observe the average costs of production at the mine level, and we seek to leverage these data to construct supply curves. Figure 6 illustrates with global data for lithium, plotting costs against cumulative capacity with mines ordered by cost. This empirical cost curve represents a global supply curve under three restrictive assumptions: myopia, linear costs, and price taking. To characterize supply, we must consider marginal revenues and costs for each mine. Under myopia, we can restrict attention to current revenues and costs. Under linear costs, marginal costs equal average costs, and so we can read marginal costs from our average cost data. Under price taking, marginal revenues equal market prices, and so we can read marginal revenues from our price data. We thus obtain a supply curve directly from the data. It predicts that mines produce to capacity when world prices exceed their observed average costs.

Myopia and linear costs are strong assumptions. Myopia is in tension with the non-renewable nature of mineral resources, which are in finite supply. Linear costs offer a local approximation that is appropriate only for evaluating small policy changes. Linear costs also imply that mines produce either at full capacity or not at all, which

is in conflict with the data. Our model relaxes both assumptions. We maintain price taking at the individual mine level, as is consistent with the large number of mines competing in these global commodity markets, although we do allow marginal revenues to diverge from market prices.

5.2 Model

We model forward-looking mines with finite mineral reserves and convex production costs, focusing our exposition on our baseline specification of costs. Appendix C.1 presents derivations, and Appendix C.2 shows that our approach generalizes to a broad class of cost specifications.

Extraction

In each period t , a mine i in country $k(i)$ extracts raw ore x^{it} . It processes this ore to produce mineral $m(i)$, which it sells in the world market. Production s^{it} depends on the grade g^i of extracted ore x^{it} .

$$s^{it} = g^i x^{it} \quad (8)$$

Marginal revenue mr^{it} is given by world price p^{mt} , price wedge τ^i , and ore grade g^i .

$$mr^{it} = (p^{mt} - \tau^i) g^i \quad (9)$$

Mines take world prices p^{mt} as given, noting that individual mines are small despite geographic concentration at the country level. Ore grade g^i is mine-specific and constant over time.⁹ Wedge τ^i is unobserved and allows marginal revenue to diverge from market prices, capturing a range of policy measures and potential distortions that include the supply restrictions we explore in counterfactuals.

⁹ In principle, ore quality may degrade over time if the highest-quality deposits are extracted first. Appendix C.3 shows how to accommodate the added dynamics from ore degradation, while also noting that ore degradation is limited in our particular empirical setting.

Cost convexity

We specify costs that are quadratic in extraction x^{it} and scale with capacity $\bar{x}^i$ through utilization rate u^{it} .

$$C^{it} = \left(\alpha^i + \frac{1}{2} \kappa^m u^{it} + \varepsilon^{it} \right) x^{it}, \quad u^{it} = \frac{x^{it}}{\bar{x}^i}, \quad (10)$$

where α^i varies by mine and κ^m by mineral. Cost level α^i captures the linear component of costs, which scales proportionally with extraction. Cost convexity κ^m captures a quadratic component of costs, which rises as mines approach or potentially exceed their extraction capacity. Unobserved cost shock ε^{it} is additively separable and linear. Appendix C.2 shows that we are not restricted to quadratic convex costs, and that we require only separability and not linearity in our cost shocks. From equation 10, it follows directly that average and marginal costs are

$$ac^{it} = \alpha^i + \frac{1}{2} \kappa^m u^{it} + \varepsilon^{it}, \quad (11)$$

$$mc^{it} = \alpha^i + \kappa^m u^{it} + \varepsilon^{it}. \quad (12)$$

If $\kappa^m > 0$, we have cost convexity with marginal costs that exceed average costs. If $\kappa^m = 0$, we have cost linearity and $ac^{it} = mc^{it}$.

Dynamics

Mines extract from finite reserves with a stock R^{it} that follows a law of motion

$$R^{it+1} = R^{it} - x^{it} \geq 0.$$

The Euler equation equates the net payoff of extracting today to its discounted expected value tomorrow.

$$mr^{it} - mc^{it} = \beta \mathbb{E}^{it} [mr^{it+1} - mc^{it+1}] \quad (13)$$

This expression is the Hotelling (1931) condition. The net payoff is the shadow cost of reserves, where $\lambda^{it} = mr^{it} - mc^{it}$. This net payoff grows at rate $1/\beta$ for discount

factor β . Substituting equations 11 and 12 and rearranging, utilization is given by

$$u^{it} = \frac{2}{\kappa^m} (mr^{it} - ac^{it} - \lambda^{it}), \quad (14)$$

reflecting observed average costs ac^{it} , convexity κ^m , and shadow costs λ^{it} . This expression nests the empirical cost curve that we describe above, in which mines operate at full capacity when marginal revenue exceeds average cost. With myopia ($\lambda^{it} = 0$), linear costs ($\kappa^m \rightarrow 0$), and binding capacity, we have $u^{it} = \mathbb{1}[mr^{it} > ac^{it}]$.

Production

We compute quantities supplied as follows. By equations 8 and 10, the quantity supplied s^{it} by a mine i in period t is given by production

$$s^{it}(\mathbf{p}^{mt}) = g^i \bar{x}^i u^{it}(\mathbf{p}^{mt}),$$

which depends on the mineral price path $\mathbf{p}^{mt} = \{p^{mt'}\}_{t' \geq t}$. For these forward-looking mines, production depends on both current and future prices. World supply aggregates across mines.

$$s^{mt}(\mathbf{p}^{mt}) = \sum_{i \in \mathcal{I}^m} s^{it}(\mathbf{p}^{mt}) \quad (15)$$

5.3 Estimation

We estimate cost parameters (α^i, κ^m) with mine-level panel data. We show that our cost data allow us to address endogeneity concerns in estimation.

Euler equation

We begin with the typical Euler approach, which derives an estimating equation from the Euler condition. We substitute equations 9 and 12 into Euler equation 13, and we take quasi-differences with notation $\Delta y^{it} = \beta y^{it} - y^{it-1}$.

$$\Delta \mathbb{E}^{it-1}[(p^{mt} - \tau^i)g^i] = \Delta \mathbb{E}^{it-1}[\alpha^i + \kappa^m u^{it} + \varepsilon^{it}]$$

We accommodate wedges of the form $\tau^i = \tau^{km} + \tau^{om}$, which include policies by both producer countries k where mines are located and owner countries o where mines'

owners are based. These policies can vary by mineral m . In expectation, the change in marginal revenue on the left-hand side is equal to the change in marginal cost on the right-hand side. We obtain an estimating equation by substituting and rewriting expected values in terms of realized values and expectational errors.

$$\Delta p^{mt} g^i = \Delta \alpha^i + \kappa^m \Delta u^{it} + \Delta \tau^{km} g^i + \Delta \tau^{om} g^i + \Delta \varepsilon^{it} + \eta^{it} \quad (16)$$

Expectational errors η^{it} capture the difference between realized and expected values.¹⁰ Positive values represent positive surprises. Under rational expectations, these expectational errors are uncorrelated with any variables contained in information set $\mathcal{N}^{it-1}$ in period $t - 1$, such that $\mathbb{E}^{it-1}[\eta^{it} | \mathcal{N}^{it-1}] = 0$ (Hall 1978).

Endogeneity

We highlight two endogeneity problems that arise from error terms $\Delta \varepsilon^{it}$ and η^{it} . First, cost shocks ε^{it} enter average costs by equation 11, and so they are correlated with utilization u^{it} by equation 14. A positive cost shock reduces utilization, and the result is downward bias in our estimate of κ^m . Second, by their definitions, difference Δu^{it} is increasing in u^{it} , while expectational error η^{it} is decreasing in u^{it} .¹¹ The result is again downward bias in our estimate of κ^m .

We address the first endogeneity problem by inverting equation 11, which allows us to express the cost shocks as a function of our data.

$$\varepsilon^{it} = ac^{it} - \frac{1}{2} \kappa^m u^{it} - \alpha^i$$

The value of our cost data is that they provide this additional moment, which we substitute into equation 16 to eliminate cost shocks ε^{it} . We obtain our baseline specification for estimation.

$$\Delta p^{mt} g^i - \Delta ac^{it} = \frac{1}{2} \kappa^m \Delta u^{it} + \Delta \tau^{km} g^i + \Delta \tau^{om} g^i + \eta^{it} \quad (17)$$

We address the second endogeneity problem by instrumenting for differenced utilization Δu^{it} with lagged utilization $u^{it-1} \in \mathcal{N}^{it-1}$. The lagged value is correlated with

¹⁰ For $\tilde{\Delta} y^{it} = \Delta y^{it} - \Delta \mathbb{E}^{it-1}[y^{it}]$, we have $\eta^{it} = \tilde{\Delta} p^{mt} g^i - \kappa^m \tilde{\Delta} u^{it} - \tilde{\Delta} \tau^i g^i - \tilde{\Delta} \varepsilon^{it}$.

¹¹ Expansion gives $\Delta u^{it} = \beta u^{it} - u^{it-1}$ and $\eta^{it} = -\beta \kappa^m u^{it} + \beta \kappa^m \mathbb{E}^{it-1}[u^{it}] + \tilde{\Delta} p^{mt} g^i - \tilde{\Delta} \tau^i g^i - \tilde{\Delta} \varepsilon^{it}$.

Δu^{it} by definition and orthogonal to η^{it} by rational expectations.

Estimation proceeds as follows. For equation 17, we construct the left-hand side from observed prices p^{mt} , grades g^i , and average costs ac^{it} , setting $\beta = 0.9$. We instrument for observed difference Δu^{it} with lagged u^{it-1} , absorbing wedges $\Delta \tau^{km}$ and $\Delta \tau^{om}$ with producer-country-mineral and owner-country-mineral fixed effects interacted with ore grade g^i . Equation 17 identifies cost convexity κ^m and wedges (τ^{km}, τ^{om}) with errors η^{it} as residuals. It does not identify cost levels α^i , which are identified instead by average cost equation 11.

$$ac^{it} - \frac{1}{2}\kappa^m u^{it} = \alpha^i + \varepsilon^{it} \quad (18)$$

Mine fixed effects isolate cost levels α^i with shocks ε^{it} as residuals. Intuitively, the $\frac{1}{2}\kappa^m u^{it}$ term corrects for cost convexity in recovering α^i from our average cost data.¹² We estimate equations 17 and 18 jointly by GMM.

Discussion

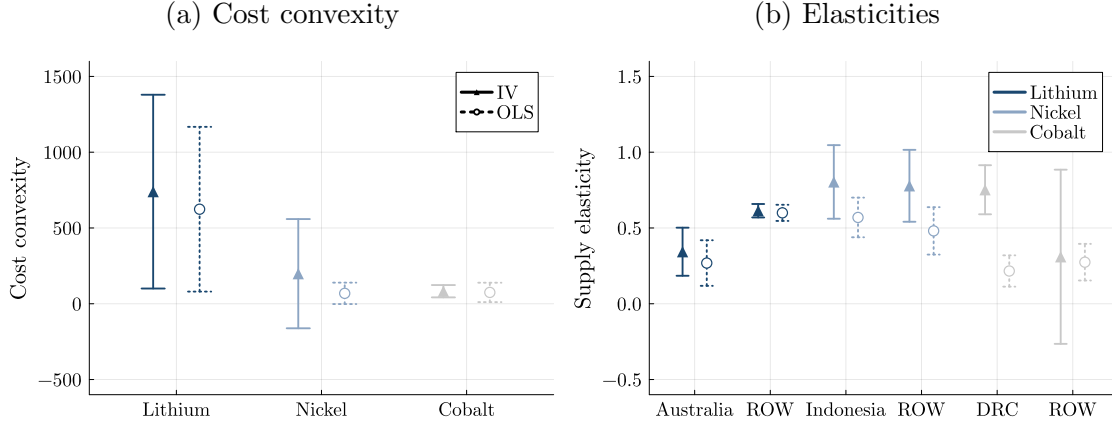
The average cost data allow us to use average cost equation 11 as an additional moment. We use this moment to eliminate cost shocks ε^{it} from estimating equation 17 and thus to address the endogeneity bias that arises from these unobserved cost shocks. Without this additional moment, we would instead require an instrument that is uncorrelated with the shocks. This approach applies both in static settings, where estimation is based on a first-order condition, and in dynamic settings, where estimation is based on an Euler equation.¹³

We draw a contrast to other approaches for leveraging cost data. [Asker et al. \(2019\)](#) assume constant marginal costs and treat average cost data as direct measures of marginal costs. This approach treats the empirical cost curve as a supply curve,

¹² By contrast, the empirical cost curve approach of Section 5.1 fixes marginal costs at the average costs observed in the data. This approach is equivalent to imposing $\kappa^m = 0$ in equation 18. Appendix C.4 shows that estimation is robust to classical measurement error in average costs.

¹³ Our dynamic setting introduces expectational error η^{it} and the need for lagged instruments. We highlight two further advantages of cost data in the dynamic setting. First, the cost moment allows us to separately identify cost shocks ε^{it} and expectational error η^{it} , as average cost equation 18 excludes the latter. We do not rely solely on Euler equation 16, which can only recover composite error $(\Delta \varepsilon^{it} + \eta^{it})$. Second, the cost moment incorporates cross-sectional variation in estimation. We do not rely solely on intertemporal Euler variation, which may be subject to noise in the exact timing of production decisions.

Figure 7: Supply estimates



We show point estimates and standard errors for cost convexity parameter κ^m and the resulting supply elasticities. We plot both IV and OLS estimates with 95% confidence intervals. For each mineral, we present elasticities for top-producing countries and the rest of the world. The elasticity is the discounted cumulative supply response over our study period in response to a permanent price increase. We compute standard errors with the delta method.

interpreting observed average costs as marginal cost levels α^i . Our framework nests this special case with $\kappa^m = 0$ and thus $ac^{it} = mc^{it}$, but we do not impose such restrictions. [Clausing et al. \(2026\)](#) build on our approach, using average cost data to estimate a static model of aluminum and steel production.

5.4 Estimates

Figure 7 reports our supply estimates. We estimate positive cost convexity κ^m , such that marginal costs rise with utilization. Furthermore, our estimates are larger with IV than with OLS, as is consistent with the downward bias from cost shocks and expectational errors that we describe above. We use the model to translate these cost estimates into supply elasticities. In particular, we simulate a permanent increase of 1% in mineral prices, and we compute the discounted cumulative response of mine production over our study period.

We report the implied supply elasticities for top producers and the rest of the world. For lithium, Australia is an established, large-scale producer that is relatively inelastic. Marginal supply comes instead from higher-cost Chinese lepidolite mines and newer African mines in the rest of the world. For nickel, production is more elastic than for other minerals, and evenly so for Indonesia and the rest of the world.

Much of the world’s nickel is found in laterite deposits that lie close to the surface, and these deposits are relatively inexpensive to mine and quick to bring online. For cobalt, the DRC has a segment of artisanal, labor-intensive mines that ramp up quickly in response to price increases, resulting in a larger elasticity than in the rest of the world. More broadly, these supply responses shape the impacts of mineral policy, which we explore in our counterfactual analysis.

6 Equilibrium

We close the model with a simple midstream layer that connects upstream mineral prices to downstream battery prices. We define an equilibrium in which mineral prices clear global mineral markets, and we describe how we solve the model for counterfactuals.

6.1 Supply chain

Recall the three broad stages of the supply chain: upstream mining and processing, midstream refining and battery manufacturing, and downstream EV manufacturing. Section 5 characterizes upstream mineral supply s^{mt} as a function of world mineral prices p^{mt} . Upstream mines produce processed minerals, which they sell to midstream refiners. Refiners produce refined minerals, which they sell to midstream battery manufacturers. Manufacturers produce batteries, which they sell to downstream EV manufacturers. Section 4 characterizes downstream battery demand d_{jt} as a function of world battery prices p_{jt} .

We now specify the midstream layer. Battery prices include mineral inputs and a midstream wedge μ_{jt} .

$$p_{jt} = \sum_{m \in \mathcal{M}} r_j^m p^{mt} + \mu_{jt}, \quad (19)$$

where μ_{jt} captures the costs of mineral refining and battery manufacturing, as well as potential markups. We recover μ_{jt} from the data as a function of observed prices (p_{jt}, p^{mt}) and recipes r_j^m . For counterfactuals, changes in mineral prices pass through to battery prices, but we hold μ_{jt} fixed. In doing so, we isolate the consequences of upstream market power and supply disruptions.

Endogenizing the midstream layer would introduce two additional channels. First, midstream costs may respond. We simulate upstream supply restrictions, which reduce midstream quantities. If midstream costs are convex, then lower quantities would reduce marginal costs and counterfactual battery prices relative to our estimates. Conversely, lower quantities would raise costs and counterfactual battery prices if midstream costs are concave or if lower production slows learning-by-doing. Second, markups may respond. Although midstream production is concentrated among large Chinese firms, these markups are potentially limited. For mineral refining, the dominant refiners are vertically integrated into battery manufacturing – either directly or through joint ventures – which may limit refining markups along internal supply chains.¹⁴ For battery manufacturing, existing estimates point to modest markups: [Barwick et al. \(2025\)](#) estimate average markups of 11% under observed market conditions.¹⁵

6.2 Market clearing

The equilibrium market-clearing condition equates mineral supply and demand for each mineral m in each period t .

$$s^{mt}(p^{mt}, p^{mt+1}, \dots, p^{mT}) = d^{mt}(\mathbf{p}^t) \quad \forall m, t, \quad (20)$$

where $\mathbf{p}^t = \{p^{mt}\}_{m \in \mathcal{M}}$ collects mineral prices. Mineral supply depends on prices over time because mines are forward-looking and have finite reserves. Mineral demand depends on prices across minerals because minerals are used jointly in battery technologies and because battery consumers can switch across technologies. Mineral demand $d^{mt}(\mathbf{p}^t)$ aggregates over battery demand $d_{jt}(\mathbf{p}_t)$, weighting by battery recipes

¹⁴ The top lithium refiner in China, Ganfeng Lithium, is vertically integrated through its subsidiary Ganfeng LiEnergy, which manufactures battery cells and energy storage systems. For cobalt, the leading midstream firms are Zhejiang Huayou Cobalt, GEM Co., and CNGR Advanced Material. They supply cathode materials directly to major battery manufacturers through joint ventures with LG Chem and POSCO. For nickel, Tsingshan Holding Group is the single largest producer at 30% of global primary nickel production, and it owns the battery manufacturer REPT Battoro.

¹⁵ [Barwick et al. \(2025\)](#) report battery markups of \$142 per kWh relative to total production costs of \$1,286 per kWh. The largest battery manufacturer reports average markups of \$83 per kWh.

r_j^m and allowing for non-EV demand ν^{mt} .

$$d^{mt}(\mathbf{p}^t) = \sum_{j \in \mathcal{J}} r_j^m d_{jt}(\mathbf{p}_t) + \nu^{mt},$$

where $\mathbf{p}_t = \{p_{jt}\}_{j \in \mathcal{J}}$ collects battery prices. Mineral supply s^{mt} is defined by equation 15, and battery demand d_{jt} by equation 6. In equilibrium, mineral prices $\mathbf{p}^t$ pin down battery prices $\mathbf{p}_t$ by equation 19.

6.3 Solving the model

We solve for the path of mineral prices $\{\mathbf{p}^t\}_{t=2024}^T$ that satisfies equilibrium condition 20 in each year until final year T . We set $T = 2035$ in our reported counterfactuals, and we explore the robustness of our results to this choice in Appendix D.2. On the supply side, we solve each mine’s optimal extraction path through year T taking the full price path as given, subject to the reserves constraint. We incorporate mine entry during this time horizon, which we capture with data on planned mine openings. We also incorporate growth in reserves, which we calibrate using observed discovery rates during our sample period. On the demand side, we construct regional battery expenditures X_{kt} from 2025 to 2035 using BMI (2025) forecasts. These forecasts allow us to incorporate current expectations over the evolution of battery technologies. For midstream wedge μ_{jt} , we recover 2024 values with equation 19. For non-EV mineral demand, we impose a linear demand function with a price elasticity of -0.1 , which we calibrate to match recent estimates of mineral demand elasticities (Shojaeddini et al. 2025). We recover the intercept from 2024 data. We set demand and supply shocks $(\varepsilon_{jkt}, \varepsilon^{it}, \eta^{it})$ to zero, such that mines face no uncertainty about the future evolution of demand or supply. Appendix D provides additional detail.

Solving over a finite horizon is consistent with our treatment of battery technologies, mine entry, and reserve discovery, which evolve exogenously in counterfactuals. Over the longer run, battery research may lead to technological innovation that reduces the demand for our minerals of interest beyond what is already captured by our forecast data. Similarly, the continued entry of new mines may reduce the price pressures from exhausting existing mines. At the same time, we note that these adjustments are likely to be slow in nature. For example, sodium-ion batteries remain in the early stages of EV deployment after decades of research, and the lead time to

establish mine production is more than 10 years (Appendix A.4).

7 Counterfactuals

We characterize market power and interdependence through three sets of counterfactual exercises. First, we assess vulnerability to upstream supply disruptions by removing the largest producer of each mineral and evaluating impacts on mineral and battery markets. Second, we simulate policies that exercise market power within mineral markets. Third, we consider policies that exercise market power across mineral markets.

7.1 Supply chain vulnerability

Concentrated mineral production makes global supply chains vulnerable to natural disaster, political instability, and other forms of disruption. We evaluate this source of vulnerability with an extreme scenario: eliminating supply from each mineral’s top producer. Australia accounts for 50% of global lithium supply, Indonesia for 53% of nickel, and the DRC for 74% of cobalt. Table 2 presents the equilibrium price effects of removing each from production.¹⁶ Lithium and nickel prices roughly double, while cobalt prices increase by over 800%. The effect for cobalt is especially pronounced, reflecting both the high baseline share of DRC production and the large ore grade advantage of DRC mines (Figure 2). When the DRC exits, other mines have low ore grades and high costs of expanding production, and so prices rise sharply.

The cross-mineral price effects reveal that lithium, nickel, and cobalt are gross complements. Lithium prices rise when we remove Australian lithium, but nickel and cobalt prices fall. As established in Section 3, this pattern is the empirical signature of gross complementarity. If these minerals were gross substitutes, then nickel and cobalt prices would rise – not fall – as rising lithium prices push consumers toward nickel and cobalt. If they were perfect substitutes, then nickel and cobalt prices would rise as much as lithium prices. Measured against this benchmark, the declines we observe are modest in absolute terms but represent a large deviation in the opposite direction, driven by the joint use of these minerals in NMC and NCA batteries. Analogous

¹⁶ We simulate permanent removal from production. Appendix D.3 evaluates temporary removals and presents price effects over time.

Table 2: Supply disruptions

	Price effects (%)				
	Lithium	Nickel	Cobalt	Battery L	Battery N
Lithium (Australia)	107	−1.09	−3.69	9.47	9.46
Nickel (Indonesia)	−1.84	76.0	−3.43	−0.17	3.55
Cobalt (DRC)	−6.19	−3.74	865	−0.58	14.1

For each mineral, we remove the top producer and compute the resulting average mineral and battery price changes over our time horizon. All effects are relative to our baseline equilibrium with no restrictive policies. Battery L is LFP, and battery N is NMC and NCA.

patterns hold for Indonesian nickel and DRC cobalt.

Battery price effects reflect battery recipes. First, battery effects are more muted than mineral effects because batteries also include non-mineral inputs, which are not directly affected by mineral supply shocks. Second, battery recipes determine the scope for substitution. When we remove Australian lithium, all battery prices rise because lithium is a common input. Consumers cannot substitute away from lithium. When we remove Indonesian nickel or DRC cobalt, prices rise for battery N (NMC and NCA) and fall for battery L (LFP). Consumers substitute toward battery L , which uses neither nickel nor cobalt, dampening the extent to which mineral price shocks pass through to battery prices.

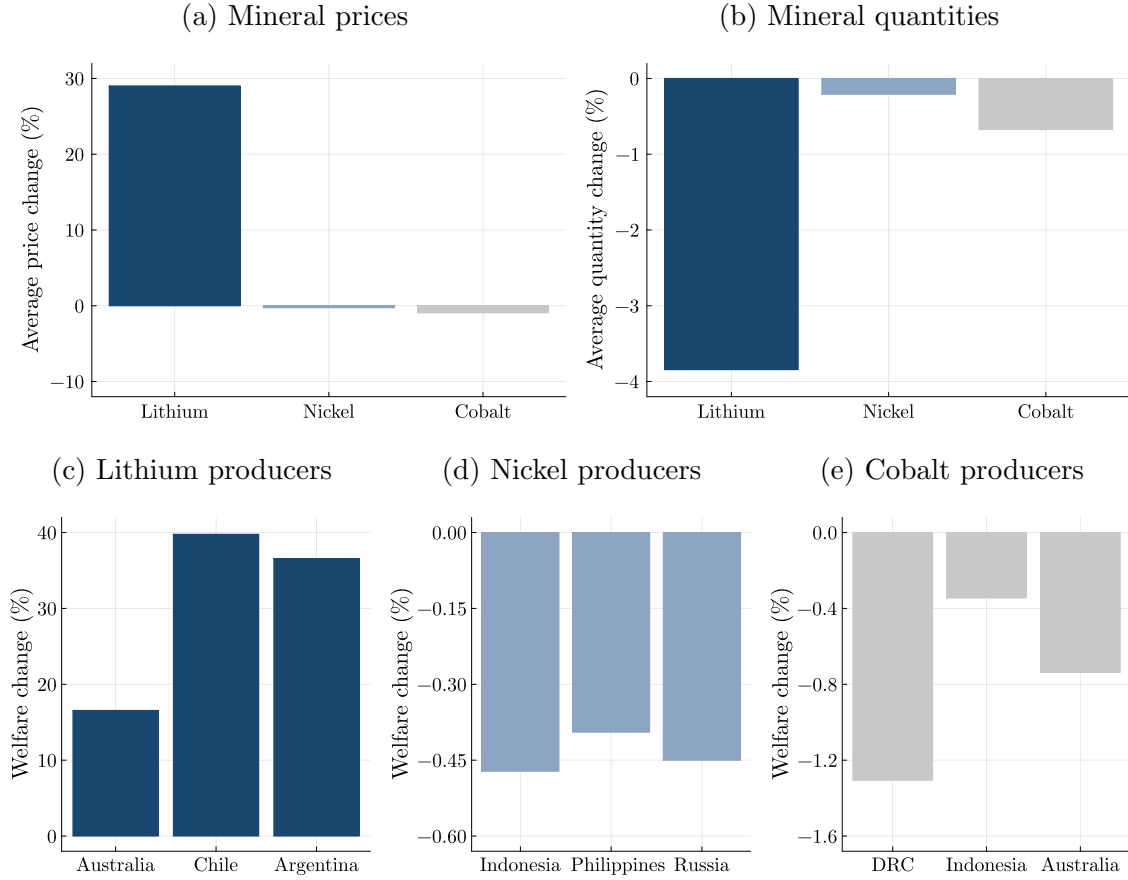
7.2 Market power within minerals

The large price effects of supply disruption suggest that mineral-producing countries have market power. We quantify this market power by allowing countries to optimally restrict supply. We discuss unilateral policy by a single country, as well as coordinated policy by a cartel of countries that produce the same mineral.

Unilateral policy

We consider a scenario in which an individual country restricts its mineral supply to exercise market power. Our main analysis simulates unilateral Australian lithium policy, which we model as a production tax set to maximize Australian welfare. We

Figure 8: Unilateral lithium policy



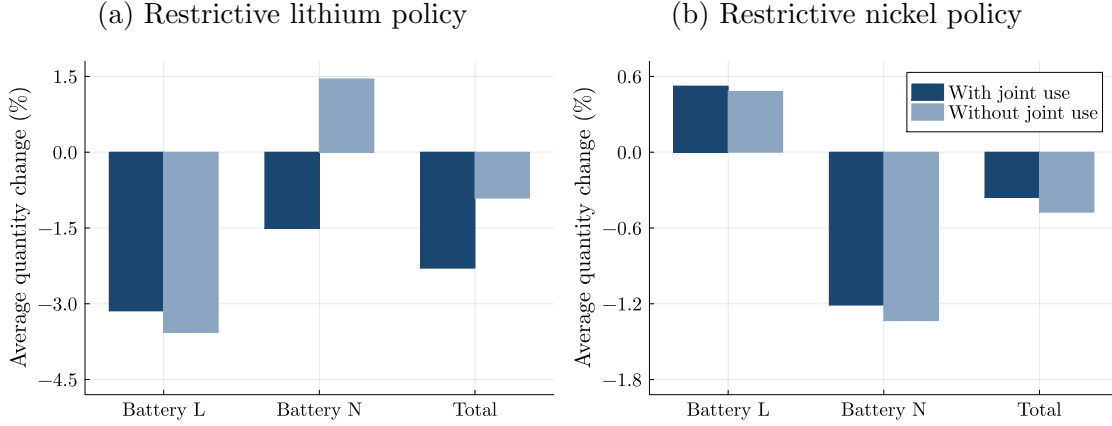
We simulate restrictive lithium policy by Australia in the form of a production tax set to maximize Australian welfare. Welfare is the discounted sum of producer surplus and government revenue over our time horizon. We report mineral market effects and welfare effects for the top three producers of each mineral. All effects are relative to our baseline equilibrium with no restrictive policies.

compute welfare as the sum of producer surplus and government revenue.¹⁷

Figure 8 presents the mineral market impacts. In panels (a) and (b), lithium prices rise and quantities fall as Australia cuts lithium production. For cobalt and nickel, prices and quantities both fall. The magnitudes are smaller than those in Table 2 because Australian policy restricts production but does not eliminate it. Nickel and cobalt prices fall rather than rise, again signaling gross complementarity. We quantify policy spillovers in panels (c), (d), and (e), which report welfare effects for top mineral producers. Lithium producers all gain, as interests are aligned among

¹⁷ The optimal production tax implements the same allocation as a centralized Australian planner seeking to maximize Australian lithium profits.

Figure 9: Green adoption and joint mineral use



We simulate restrictive lithium policy by Australia in the form of a production tax set to maximize Australian welfare, and we similarly simulate restrictive nickel policy by Indonesia. Welfare is the discounted sum of producer surplus and government revenue over our time horizon. We report battery quantity effects both with and without the joint use of minerals in battery production. All changes are relative to our baseline equilibrium with no restrictive policies. Battery *L* is LFP, and battery *N* is NMC and NCA.

substitute minerals. Australia gains from its own policy, exercising market power by restricting lithium supply to raise lithium prices. Chile and Argentina gain even more than Australia, free-riding on high lithium prices without restricting their own supply. By contrast, nickel and cobalt producers all lose. Interests are in conflict across complementary minerals.

Figure 9 shows green adoption outcomes under Australian lithium policy and Indonesian nickel policy. We simulate each with and without the joint use of minerals in battery production.¹⁸ For restrictive lithium policy, joint use leads to larger reductions in green adoption. The reason is that lithium is a common input in the production of both batteries. Without joint use, lithium is no longer a common input, and so consumers can substitute away from lithium by shifting toward battery *N*. The resulting rise in battery *N* adoption partially offsets the initial fall in battery *L* adoption. With joint use, lithium is a common input that affects the prices of both batteries, and so consumers cannot substitute away from lithium by switching batteries. The fall in total green adoption more than doubles.

¹⁸ We shut down joint use by removing lithium from battery *N* production, such that lithium is exclusive to battery *L*. Nickel and cobalt remain exclusive to battery *N*. Appendix D.2 shows that our results are robust to alternative specifications for shutting down joint use.

For restrictive nickel policy, joint use leads to smaller reductions in green adoption. Nickel is not a common input, either with or without joint use, and so consumers can always substitute toward battery L and away from nickel. Joint use instead captures complementarity. With joint use, because battery N also uses lithium, nickel restrictions reduce lithium demand and lithium prices. Lower lithium prices subsidize battery L adoption, and the fall in total green adoption is attenuated. These experiments demonstrate and quantify the theory in Section 3: the common input force is large and hurts adoption, while the complementarity force is smaller and helps adoption.

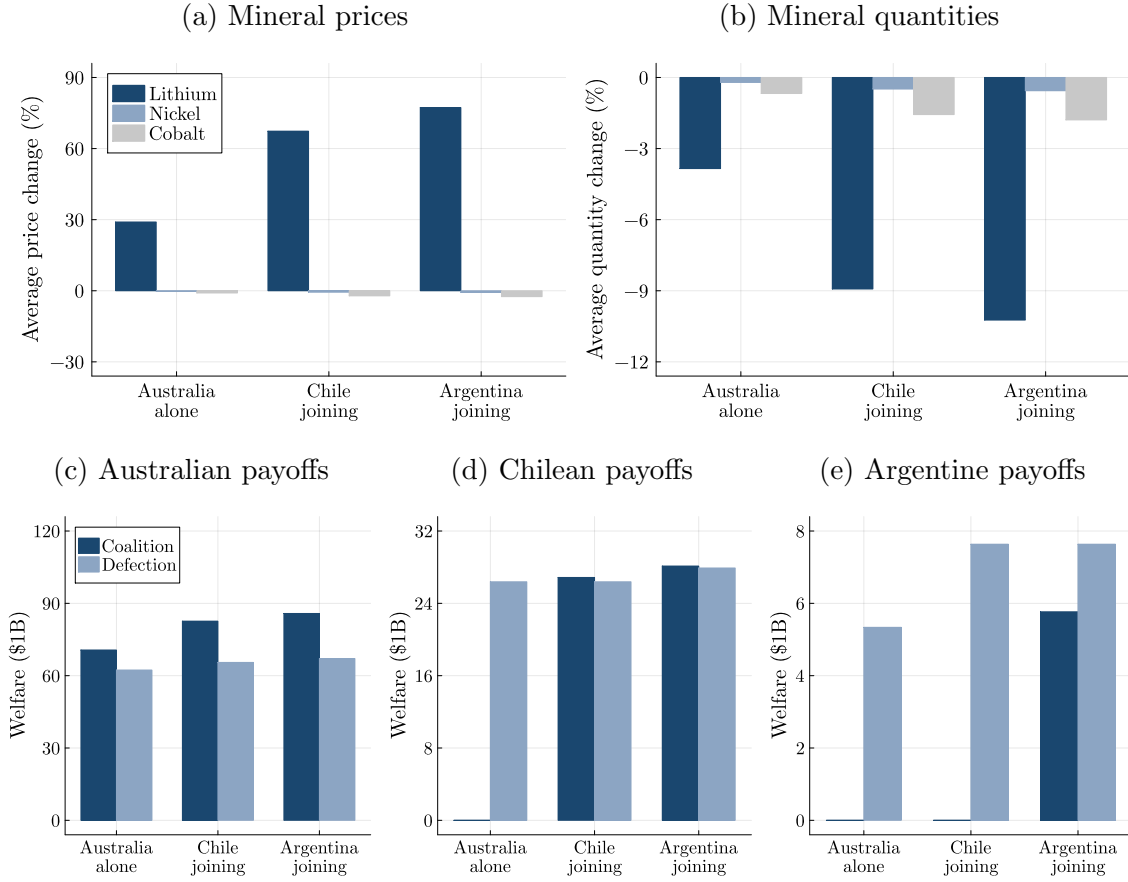
Mineral cartels

We turn to policies set by countries that form a cartel within a single mineral market. This cartel of substitutes raises standard collusive concerns, which are familiar from OPEC in oil markets. We simulate a cartel of the top lithium producers: cartel members impose a joint tax on their production to restrict supply and maximize cartel surplus, which we compute as collective producer surplus and tax revenue. Figure 10 shows price and quantity effects as the coalition expands from the largest producer to the top three producers. Impacts intensify as the lithium cartel grows. Australia acting alone raises prices by 30% and reduces quantities by 4%. These magnitudes more than double as Chile joins Australia, while Argentina has a smaller marginal effect. The cartel’s power lies with its two largest members.

Larger cartels may be more consequential but less cohesive. For each coalition, we report each member’s welfare both in coalition and when it defects.¹⁹ Australia always prefers the cartel, and its gains increase as the cartel grows. Australia anchors the coalition. Chile is roughly indifferent between staying and defecting, whether it joins Australia alone or alongside Argentina. Small transfers from Australia can sustain Chilean participation, with major consequences for market outcomes. Argentina prefers to defect, and it gains an additional \$2 billion by free-riding on Australian and Chilean supply cuts. Free-riding incentives again call for transfers, although \$2 billion is sizable relative to the cartel’s gains from Argentine participation.

¹⁹ A stable cartel requires (1) that all cartel members are better off being in a cartel than not and (2) that members do not find it profitable to cheat ([Asker and Nocke 2021](#)). Our exercise explores the first requirement, taking as given the second requirement of enforcement within the cartel.

Figure 10: Lithium cartels

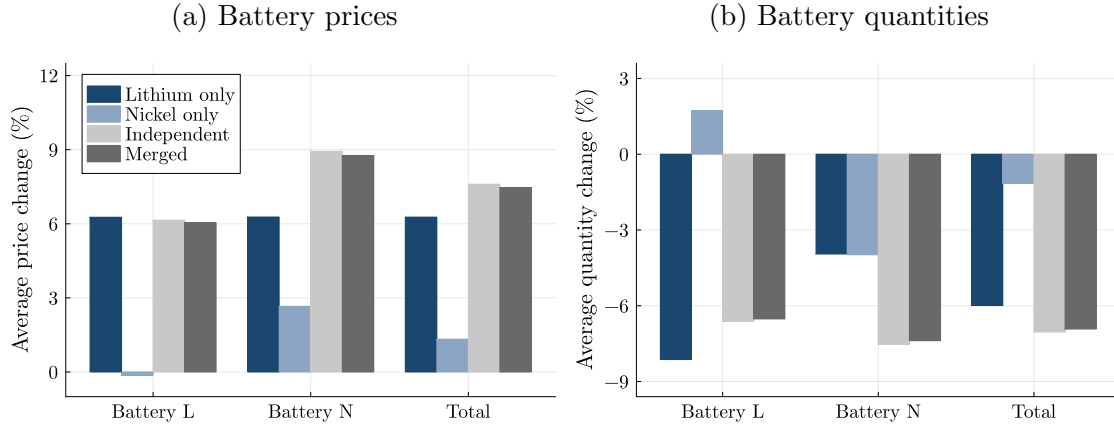


We simulate a lithium cartel that implements restrictive policy in the form of production taxes, which are set to maximize collective cartel welfare. Welfare is the discounted sum of producer surplus and government revenue over our time horizon. We consider three coalitions: Australia alone; Australia and Chile; and Australia, Chile, and Argentina. All effects are relative to our baseline equilibrium with no restrictive policies.

7.3 Market power across minerals

Market power extends and interacts across minerals because battery production uses minerals jointly. We quantify the market power of multi-mineral cartels and of multinational firms with mining stakes across minerals. We then contrast the two through the lens of resource nationalism.

Figure 11: Lithium and nickel cartels



We simulate four scenarios: a lithium cartel alone (as in Figure 10), a nickel cartel alone, a lithium cartel and a nickel cartel that operate independently, and a merged lithium-nickel cartel that operates jointly. All involve the top three producers for a given mineral and set production taxes to maximize collective cartel welfare. Welfare is the discounted sum of producer surplus and government revenue over our time horizon. All effects are relative to our baseline equilibrium with no restrictive policies. Battery *L* is LFP, and battery *N* is NMC and NCA.

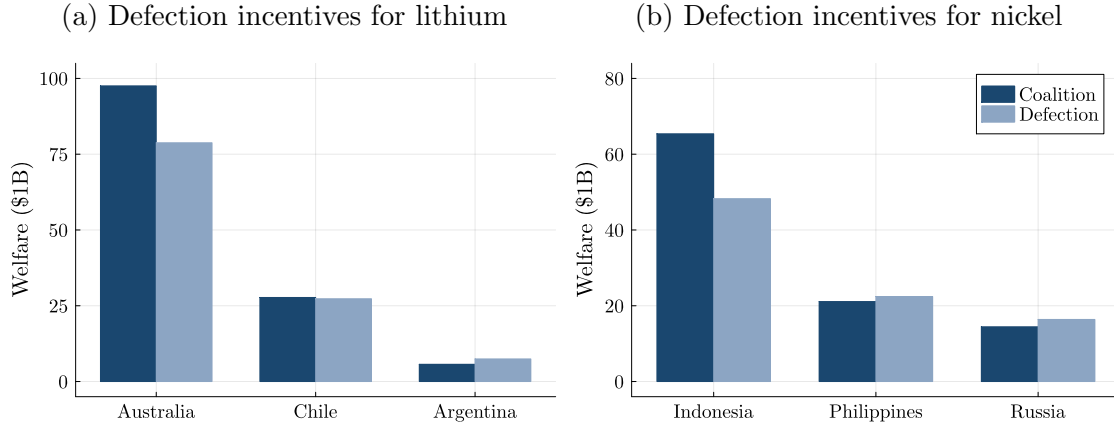
Multi-mineral cartels

We study a multi-mineral cartel of the top lithium and nickel producers. Consider a lithium cartel that includes Australia, Chile, and Argentina and a nickel cartel that includes Indonesia, the Philippines, and Russia. We quantify the effects of each cartel in turn. We then suppose that both cartels act independently, maximizing own-cartel welfare. Lastly, we suppose that the two cartels merge into a single cartel, which maximizes collective cartel welfare.

Figure 11 reports battery market effects. When the lithium and nickel cartels each act alone, they restrict supply and raise battery prices. Green adoption falls. The lithium cartel has larger effects because lithium is a common input to both batteries, while the effects of the nickel cartel are concentrated in battery *N*. When the cartels operate simultaneously but independently, we have consolidation within both mineral markets. The simultaneous operation of both cartels is unambiguously anti-competitive, and its effects are larger than those of either cartel acting alone.

When the cartels merge and operate jointly, we also have consolidation across mineral markets. This further consolidation introduces two competing forces. On one hand, the merged cartel internalizes that lithium and nickel are substitutes across

Figure 12: Lithium-nickel cartel cohesion



We simulate restrictive lithium and nickel policies in the form of production taxes set to maximize collective cartel welfare. Welfare is the discounted sum of producer surplus and government revenue over our time horizon. We plot each country's welfare gains from remaining in the cartel coalition and from defecting. All effects are relative to our baseline equilibrium with no restrictive policies.

batteries, which strengthens the incentive to restrict output – the standard anti-competitive force. On the other hand, the merged cartel internalizes that lithium and nickel are complements within battery N , which instead weakens the incentive to restrict output – the pro-competitive force of Cournot complementarity (Cournot 1838). We find that the two forces offset each other, leaving the merged cartel's impact close to that of the independent cartels. If anything, complementarity modestly dominates, as is consistent with our finding that minerals are gross complements. On net, the cartel merger has small pro-competitive effects.

The cohesion of the merged cartel depends on the distribution of cartel gains. We compute each member's welfare within the merged cartel and when it unilaterally defects to free-ride on the cartel's supply restrictions. The difference between defection and remaining in coalition captures the incentive to defect, which we plot for each member in Figure 12. The two dominant producers, Australia and Indonesia, remain firmly anchored: each would lose nearly \$20 billion of net present value by defecting. Chile participates without the need for transfers, although only weakly so. Argentina, the Philippines, and Russia each gain from defection, and these gains undercut cartel cohesion.

Multinational ownership

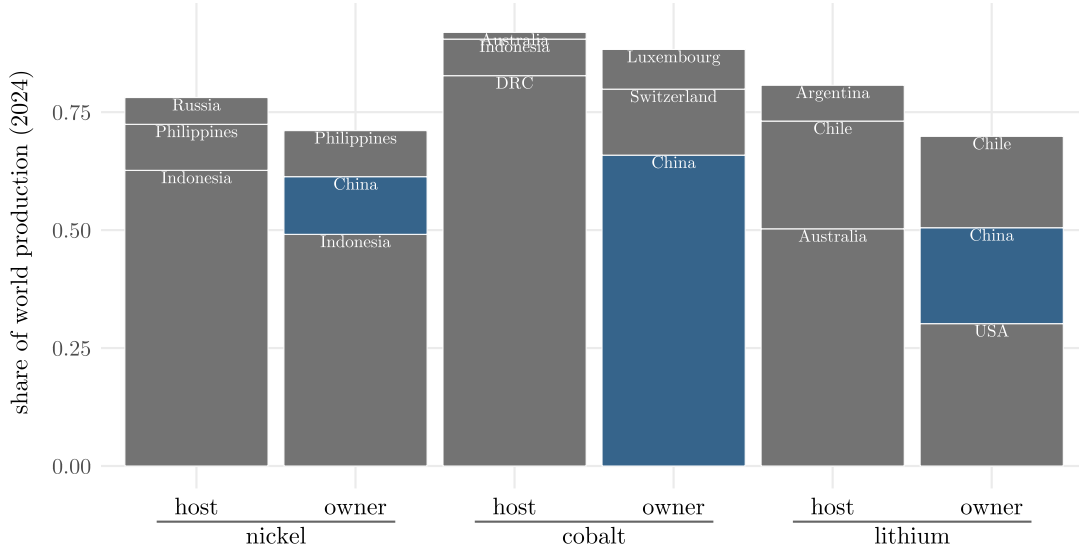
Multi-mineral cartels point to a broader theme of horizontal coordination across mineral markets. Multi-mineral cartels coordinate at the country level, but such coordination can also occur at the firm level through multinational, multi-mineral ownership. Indeed, while mineral resources are geographically concentrated, mines are frequently owned by foreign firms.²⁰ Figure 13 shows how market concentration changes when production is assigned to the country where the mine owner is headquartered, rather than the country where the mine is physically located. China is the only country with multinationals that hold significant stakes across all mineral markets. This mine ownership is a potential source of market power.

Figure 14 shows the impacts of China’s upstream market power, which it exercises through its multinationals. First, we simulate optimal Chinese policy for each mineral independently, without coordinating across mineral markets. Second, we repeat the exercise allowing for horizontal coordination across mineral markets. We find that prices fall for all minerals under horizontal coordination, relative to uncoordinated policy, as China internalizes complementarity within its multi-mineral portfolio of mines. Chinese ownership shares are modest for some minerals, limiting the magnitude of this effect, but the direction is consistent with our finding that multi-mineral mergers can be pro-competitive. Complementarity softens the impacts of mineral market power.

China is also a major battery consumer. We simulate vertical coordination by allowing China’s upstream mineral policy to account for its downstream battery consumption. Using our battery demand system, we compute downstream welfare as the consumer surplus of Chinese EV manufacturers. Figure 14 shows that China sets less restrictive policy once it internalizes that high mineral prices reduce downstream consumer surplus. Concentrated upstream ownership creates the capacity to exercise market power, but vertical linkages can discipline the incentive to restrict mineral supply.

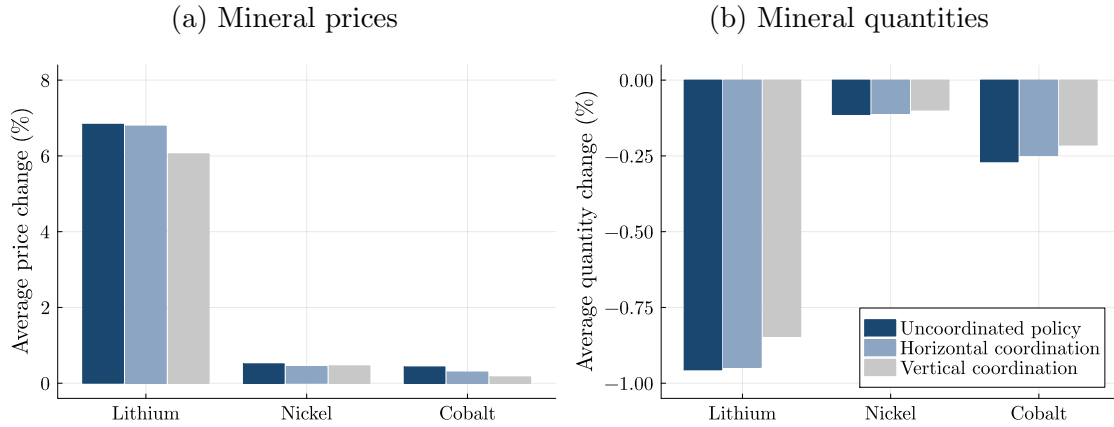
²⁰ Appendix D.3 shows how foreign ownership may influence mineral policy, focusing on the case of Australian lithium.

Figure 13: Market concentration by mine ownership



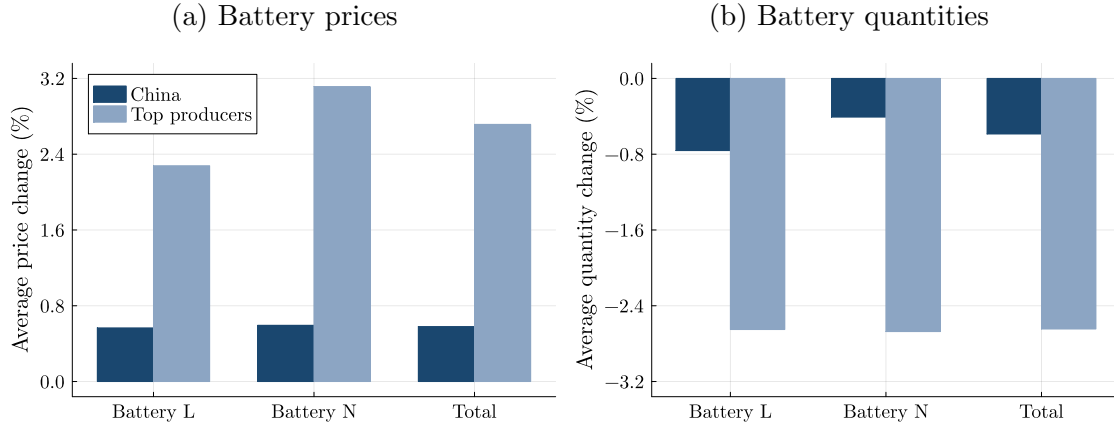
Host country is defined as the country of the mine's physical location. Owner country is defined as the country in which the mine's owner is headquartered. Owner country shares are computed by assigning a mine's production to its owning firms using equity shares. Firm-level production is then assigned to the country where its headquarters are located.

Figure 14: Chinese policy



We simulate three scenarios in which China sets lithium, nickel, and cobalt production taxes on Chinese-owned mines. For uncoordinated policy, we set taxes independently across minerals to maximize upstream Chinese welfare within each mineral market. Welfare is the discounted sum of producer surplus and government revenue over our time horizon. For horizontal coordination, we set taxes jointly across minerals to maximize upstream Chinese welfare. For vertical coordination, we set taxes to maximize upstream Chinese welfare, as well as downstream Chinese welfare. All changes are relative to our baseline equilibrium with no restrictive policies.

Figure 15: Chinese multinationals vs. top mineral producers



We simulate a scenario in which Australia (the top lithium producer) and Indonesia (the top nickel producer) independently set production taxes to maximize their individual welfare. We compare it to a scenario in which Chinese-owned lithium and nickel mines jointly set production taxes to maximize their total welfare. Welfare is the discounted sum of producer surplus and government revenue over our time horizon. All changes are relative to our baseline equilibrium with no restrictive policies. Battery *L* is LFP, and battery *N* is NMC and NCA.

Resource nationalism

Market power in natural resources can be exercised by multinational firms or resource-endowed countries, as emphasized above. Moreover, the balance of market power between the two shifts over time. These shifts are in part shaped by geopolitical events, as suggested by historical episodes of resource nationalization in oil markets (Yergin 2012). In particular, oil nationalization episodes of the 20th century transferred control of oil assets from Western multinationals to Middle Eastern governments, replacing the market power of multinationals with that of OPEC. Critical minerals differ from oil because multinational firms can coordinate across complementary minerals, while sovereign states typically exercise market power within individual mineral markets. We evaluate the implications of this distinction for resource nationalism by comparing the market power of Chinese multinationals to that of the top mineral producers.

Figure 15 presents changes in battery prices and quantities for two scenarios. In the first scenario, market power is exercised independently by the top lithium and nickel producers. In the second scenario, China exercises market power across the lithium and nickel mines owned by its multinationals. Battery prices rise and

quantities fall in both scenarios as policymakers exercise market power, but these effects are larger in the top-producers scenario. Two mechanisms drive this result. The first is simply that Chinese market shares are lower in our data. The second is that Chinese policy internalizes complementarity across minerals, while the top mineral producers do not. Each pushes toward less restrictive policy. Based on current Chinese ownership shares, resource nationalization may amplify market power, raising battery prices and slowing green adoption.

8 Conclusion

We argue that the structure of critical mineral markets fundamentally shapes the pace and pattern of the green transition. Critical minerals are highly concentrated in a small number of countries, and these countries can exercise market power by restricting supply. We show that the joint use of critical minerals determines how mineral concentration affects battery adoption, both in sign and in magnitude. For restrictive lithium policy, lithium’s role as a common battery input prevents consumers from substituting away, deepening the fall in green adoption. For restrictive nickel policy, complementarity causes lithium prices to fall as nickel supply contracts. Lower lithium prices subsidize lithium-intensive batteries, attenuating the fall in green adoption. Coordinated cartel policy combines both aspects of joint use: extending market power over common inputs deepens the fall in green adoption, while internalizing complementarities attenuates it. Taken together, these results show that mineral market power in these concentrated markets cannot be understood mineral by mineral. We emphasize the interconnected nature of these key inputs for green technology. Our climate progress will depend on these critical markets.

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APPENDIX

A Data

We construct our mine-level panel in three stages. First, we link mine records across BMI, GlobalData, and S&P Global, which lack common identifiers and use inconsistent mine names. Second, we standardize output measures across deposit types, commodity formats, and reporting units. Third, we validate the resulting panel against independent USGS production aggregates. We then present our battery recipes, along with additional descriptive evidence.

A.1 Mine-level panel data

Linking mine records. We combine data from three proprietary sources – [BMI \(2025\)](#), [GlobalData \(2025\)](#), and [S&P Global \(2025\)](#) – to create a mine-level panel for lithium, nickel, and cobalt from 2010 to 2024. There are no common identifiers across datasets, and mine names are not always standardized. We therefore construct a mine-level crosswalk using names, locations, ownership information, production, and capacity, and we verify each match individually. Our final panel consists of 406 mines that produce lithium, nickel, or cobalt.

Standardizing output measures. Mine output is reported in multiple commodity formats that vary with deposit type. Because these formats contain different amounts of the underlying mineral, the reported quantities are not directly comparable. We standardize these output measures by converting quantities to a common unit of account for each mineral.

Lithium occurs in two types of deposits: liquid brine and hard rock. South American lithium is typically found as underground brine, which is pumped up to the surface into large pools, evaporated by sunlight, concentrated, and then precipitated to obtain lithium concentrate. It is common for lithium carbonate to be produced directly on site from the concentrate. Australian lithium occurs mainly in hard-rock, spodumene-bearing pegmatites. Raw ore is extracted and treated with chemicals to extract spodumene concentrate. Given the prevalence of South America and Australia in lithium mining, lithium concentrates are typically traded internationally as lithium

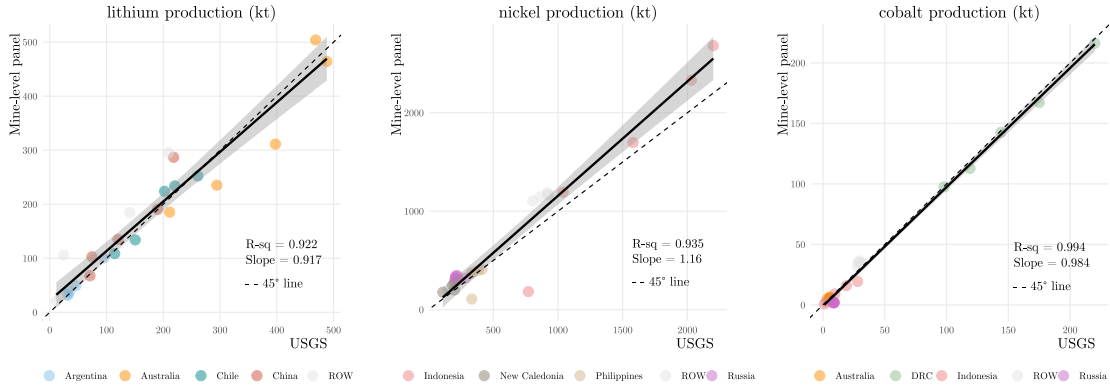
carbonate or spodumene concentrate. Lithium carbonate can be processed further into lithium hydroxide, while spodumene concentrate can be processed into either lithium carbonate or lithium hydroxide. We follow the industry standard of reporting lithium output in units of lithium carbonate equivalent (LCE), and we convert all lithium quantities to tonnes of LCE.

Nickel occurs in two types of ore: laterite and sulfide. Laterite ores are typically located near the surface and mined from open pits. They are found in Indonesia, the Philippines, and New Caledonia. Sulfide ores are mined deep underground and are found in Russia, Canada, and Australia. Nickel ore can be exported directly in raw form or in processed form. Among the processed nickel products, there are two broad categories: “Class 1” contains at least 99.8% nickel, while “Class 2” contains less than 99.8%. Class 1 nickel is primarily produced from sulfide deposits and traded on the London Metal Exchange (LME), the global market for most non-ferrous metals. Class 2 nickel comprises lower-grade nickel products such as ferronickel, nickel oxides, and nickel pig iron. Beyond these two classes, nickel supply chains include a range of intermediate nickel products that typically contain 40 to 70% nickel. These intermediate products are increasingly used to produce nickel sulfate, a chemical compound used in some lithium-ion batteries. Examples include mixed hydroxide precipitate (MHP), mixed sulfide precipitate (MSP), nickel hydroxide cake, and nickel matte. We convert all nickel quantities to tonnes of elemental nickel.

Cobalt is typically found in hard-rock deposits, including in the Central African Copperbelt. Once extracted, cobalt is often processed on site through leaching to produce cobalt hydroxide. This cobalt hydroxide is the typical export from the DRC to China, where refiners use it to produce cobalt sulfate. The cobalt sulfate is then sold to battery cathode producers. These refiners may also produce cobalt oxide in the process, which is sold for non-battery applications that include ceramics and pigments. We convert all cobalt quantities to tonnes of elemental cobalt.

Validating the data. We perform several validation exercises to gauge the coverage and representativeness of our harmonized panel. We use USGS values for 2020 to 2024 as reference values for world and country-level production. First, our mines cover 94% of world production for lithium, 94% for cobalt, and 79% for nickel. Second, we aggregate our mine-level data to the country level and compare our aggregated

Figure A1: Validation of mine-level panel against USGS production values



The vertical axes display country-level production values aggregated from our mine-level data. The horizontal axes show country-level production values reported by USGS for 2020 to 2024.

values to those from USGS, which does not report production below the country level. Figure A1 shows that our measures are highly correlated with USGS values. All R -squared values exceed 0.9.

A.2 Battery recipes

Battery recipes come from BatPaC, developed by Argonne National Laboratory (Knehr et al. 2024b). Their lithium-ion battery cost model includes the mineral intensities for a large number of EV battery technologies. Figure A2 summarizes these recipes. While many values are pulled directly from the BatPaC model, a few require additional explanation. First, aluminum enters some batteries, including NCA, as an active cathode material, and it enters all batteries as part of the current collector foil. We include both uses of aluminum. Second, we convert quantities of iron phosphate (FePO_4) to quantities of phosphoric acid (H_3PO_4), a precursor to iron phosphate. The chemical reaction is $\text{Fe}_2\text{O}_3 + 2 \text{H}_3\text{PO}_4 \longrightarrow 2 \text{FePO}_4 + 3 \text{H}_2\text{O}$. Since FePO_4 is 150.8 g/mol and H_3PO_4 is 98.0 g/mol, one tonne of iron phosphate requires 98.0/150.8 tonnes of phosphoric acid. Third, we aggregate NMC varieties to a representative NMC category by averaging over the recipes, weighting by market shares.

Figure A2: Mineral intensities (kg/kWh) across battery technologies

	LFP	NCA	NMC-111	NMC-523	NMC-622	NMC-811
phosphorus	0.38	-	-	-	-	-
phosphoric acid	1.43	-	-	-	-	-
nickel	-	0.65	0.32	0.43	0.51	0.61
manganese	-	-	0.3	0.24	0.16	0.07
lithium	0.09	0.1	0.12	0.11	0.1	0.09
iron phosphate	2.2	-	-	-	-	-
iron	0.68	-	-	-	-	-
graphite	0.95	0.85	0.84	0.84	0.85	0.85
fluorine	0.09	0.06	0.07	0.06	0.06	0.06
copper	0.62	0.33	0.39	0.36	0.36	0.32
cobalt	-	0.12	0.32	0.17	0.17	0.08
aluminum	0.24	0.14	0.15	0.14	0.13	0.12

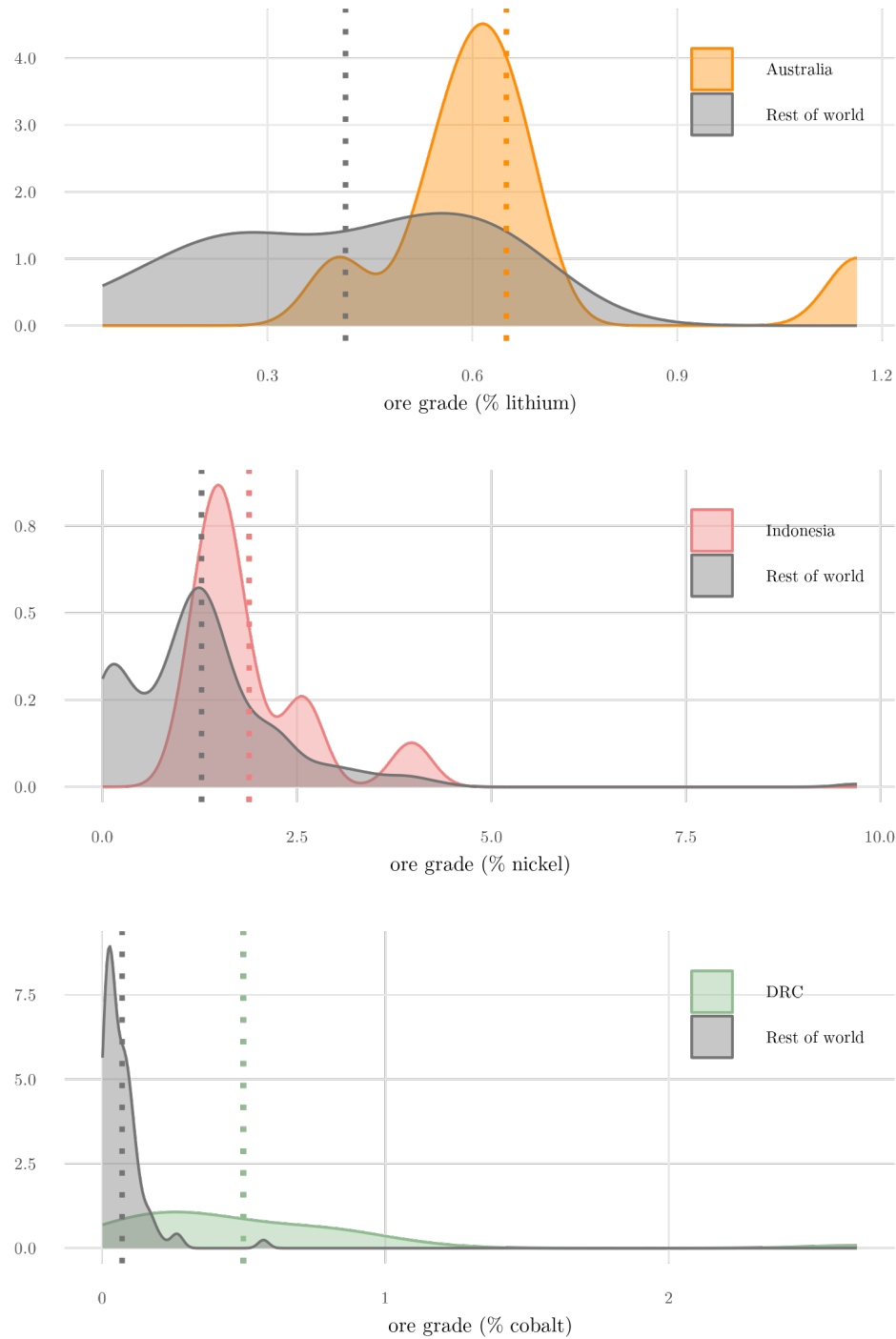
Mineral intensities are obtained from the battery technology data described in Section 2.2.

A.3 Mine heterogeneity

Ore grade. Ore grade is a major determinant of a mine’s profitability, and it explains why some countries dominate specific mineral markets. Figure A3 shows the distribution of ore grades across all the mines in our data, separating those in the top-producing country of each mineral and those in the rest of the world. While Indonesian nickel grades are higher than those in the rest of the world, the gap is not as large as that for DRC cobalt. The DRC’s significantly higher-grade cobalt deposits are a major reason why it dominates the global cobalt market.

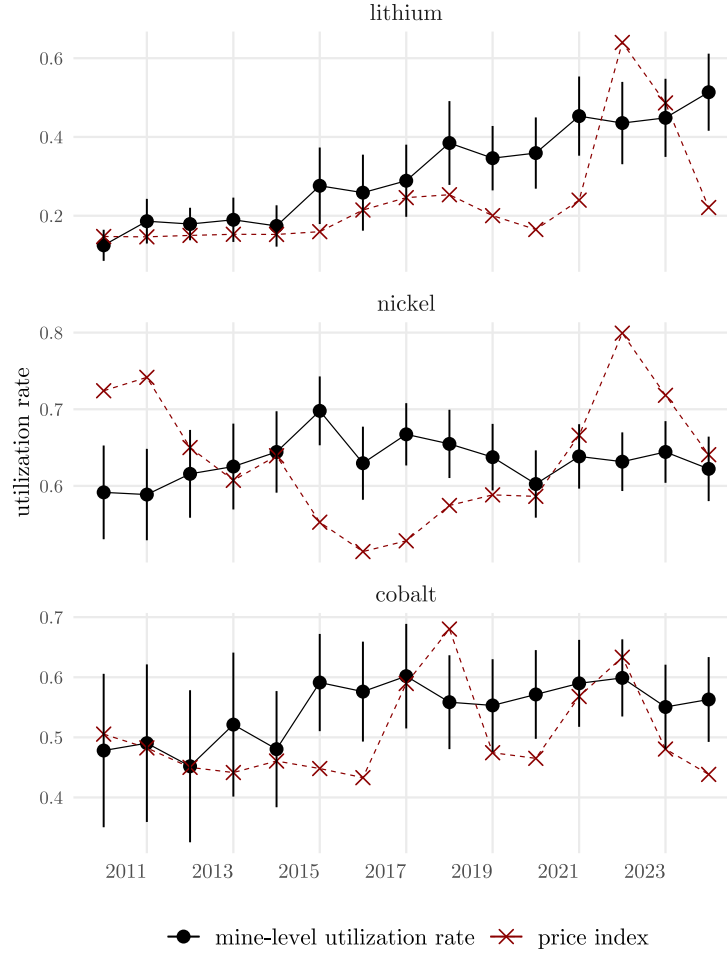
Capacity and utilization rates. Figure A4 shows how mine-level utilization rates change over time and how they correlate with world prices. Because mine owners are

Figure A3: Distribution of ore grade across mines



The figure shows density plots constructed from mine-level ore grade data described in Section 2.2. The distribution of mines from the top producer of each mineral is compared to mines in the rest of the world. Vertical lines show the average ore grade within each region.

Figure A4: Mine-level utilization rates and price fluctuations



All values are constructed from mine-level production and capacity data described in Section 2.2. The distribution of mine-level utilization is displayed in black with a 95% confidence interval. World commodity prices are reported as an index for visualization.

forward-looking, utilization does not necessarily respond to contemporaneous price spikes. Instead, production decisions are based on both current and future expected prices. We use a dynamic mining model to estimate mineral supply elasticities from observed variation in prices and utilization.

A.4 Mine entry

We treat mine entry as exogenous in our empirical model. Consistent with this assumption, mining projects have notoriously long development times. The time from

discovery to production is measured in decades, with lead times in most countries that range from 15 to 25 years (Bonakdarpour et al. 2024). The United States has the world’s second-longest average lead time of 29 years, partly due to uncoordinated and overlapping regulatory jurisdictions on the federal lands where most mining resources are located. Zambia has the world’s longest lead time of 34 years. By contrast, conventional oil wells have lead times of roughly 5 years (Wachtmeister and Höök 2020). One reason for long lead times is that mining is more environmentally disruptive than other extractive activities. Mining often requires stripping entire swaths of land, while drilling operations largely take place underground or offshore. Mining projects are thus subject to extensive exploration and feasibility studies, lengthy permitting processes, and higher litigation risk from local opposition.

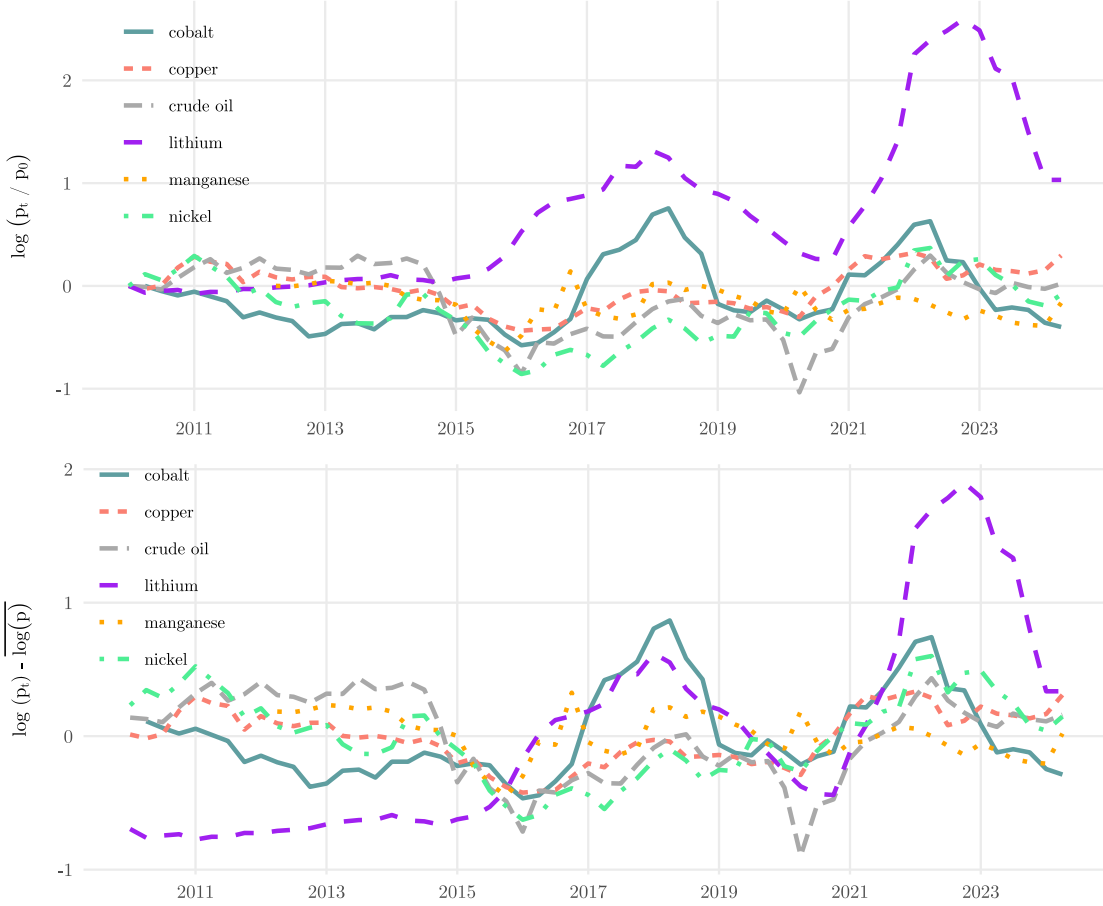
A.5 Mineral market trends

From 2017 to 2022, demand from the energy sector was the main factor behind a tripling in global production of lithium and a 70% increase in cobalt production. During this period, the EV share of mineral demand for lithium, cobalt, and nickel grew from 15% to 60%, 15% to 30%, and 2% to 10% (IEA 2023). We connect trends in battery demand downstream to fluctuations in mineral prices upstream.

Lithium. The increase in EV demand drove lithium prices to rise through 2017 (Figure A5). Lithium mines in Australia, Argentina, and Chile responded by expanding production, alleviating price pressures by 2019. Demand began to surge in 2021, and lithium prices reached their highest peak by the end of 2022. High prices again encouraged mines to expand production, and lithium prices returned to their 2021 levels by late 2023.

Nickel. In 2014, Indonesia implemented a ban on unprocessed nickel ore exports to encourage domestic smelting. Prices initially rose, but gradually fell by 2016 as alternative production sources and Indonesian domestic smelters came online (Figure A5). In February 2022, fears of sanctions on Russian nickel led to price increases. Additionally, in March 2022, a short squeeze on the London Metal Exchange caused nickel prices to surge further. Nickel prices reached a peak in 2022, encouraging demand substitution toward battery technologies that were less reliant on nickel, including LFP. This substitution contributed to the subsequent decline in nickel prices.

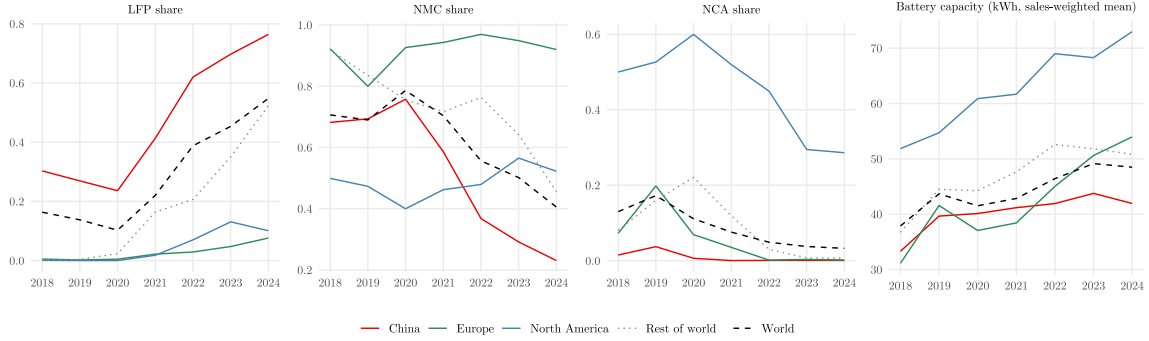
Figure A5: Mineral price trends and cycles (2010–2024)



The two panels show the price of each commodity normalized by (i) its price at the start of the sample, and (ii) its average value across the sample period. Nominal values from which this figure is constructed are quarterly averaged prices from [S&P Global \(2025\)](#). Cobalt, nickel, and copper prices are based on over-the-counter (OTC) and contract for difference (CFD) financial instruments, primarily from the LME. Crude oil prices are WTI futures. Lithium prices are spot prices for lithium carbonate (battery grade, 99.5%) traded in China. Manganese prices are Tianjin port prices.

Cobalt. Rapid growth in demand from the EV sector partly explains the rise in cobalt prices between 2016 and early 2018 (Figure A5). Disruptions by major suppliers also fueled fears of supply security. Manufacturers began shifting to batteries with lower cobalt intensities in response to cobalt price spikes and ethical-sourcing concerns. Such shifts contributed to the price decline after 2018.

Figure A6: Regional market shares of EV battery technologies



Market shares constructed from units sold, using EV sales data from [BMI \(2025\)](#).

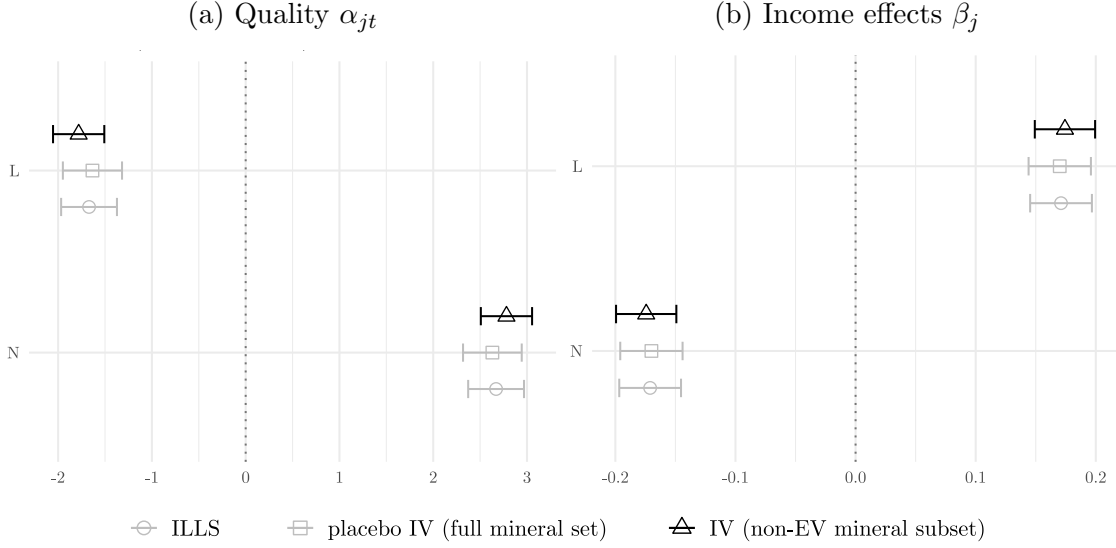
A.6 Battery market trends

Figure A6 shows the evolution of EV battery technologies worldwide and across four consumer regions: China, Europe, North America, and the rest of the world. As of 2024, consumers in China accounted for over 60% of global EV sales, while Europe and North America represented 20% and 10% of the market. Beginning in 2018, every region had positive NMC and NCA shares, while LFP was less than 1% everywhere except in China. Since then, LFP has increased its market share globally, especially in China and the rest of Asia, primarily at the expense of NMC. In Europe and North America, NMC remains the main battery technology. NCA has fallen to near zero market share in all regions except North America, where it is still used by premium brands for its high energy density. The far-right panel of Figure A6 plots the average battery capacity of EVs sold in each region.

B Demand

Estimation. We apply the standard adding-up, homogeneity, and symmetry restrictions in estimating demand. Adding-up imposes $\sum_j w_{jkt} = 1$ for all k, t , and it implies that $\sum_j \alpha_{jt} = 1$, $\sum_j \beta_j = 0$, and $\sum_j \gamma_{jj'} = 0$ for all j' . Homogeneity imposes $\sum_{j'} \gamma_{jj'} = 0$ for all j . Symmetry imposes $\gamma_{jj'} = \gamma_{j'j}$ for all j, j' . The adding-up restriction results in collinearity among the J expenditure share equations, and so we drop one of these equations in estimation. With two batteries L and N , we drop

Figure B1: Demand estimates of quality and income effects



Vertical axes label the subindices of each parameter. We report median quality α_{jt} over time. We cluster standard errors by quarter, allowing for serial correlation with a lag of one quarter.

the expenditure share equation for battery N , and estimation proceeds with a single equation for battery L . We further impose that quality α_{Lt} varies over time only through a common linear time trend, such that $\alpha_{Lt} = \alpha_{L0} + \alpha_{L1}t$. Applying these restrictions, we obtain our estimating equation.

$$w_{Lkt} = \alpha_{L0} + \alpha_{L1}t + \beta_L \log \left(\frac{X_{kt}}{P_t} \right) + \gamma_{LL} \log \left(\frac{p_{Lt}}{p_{Nt}} \right) + \varepsilon_{Lkt}$$

Relative prices $\frac{p_{Lt}}{p_{Nt}}$ appear on the right-hand side because, by homogeneity, we have $\gamma_{LL} = -\gamma_{LN}$.

Quality and income effects. Figure B1 reports estimates of quality α_{jt} and income effects β_j . Panel (a) shows lower estimated quality α_{jt} for battery L , rationalizing lower shares despite lower prices in the data. Panel (b) shows larger income effects β_j for L , reflecting that rising aggregate battery expenditure coincides with the rapid growth of LFP adoption during our sample period.

Expenditures. We hold nominal total battery expenditures X_{kt} fixed in our counterfactuals. Adjusting for battery price index P_t , we obtain real expenditures $Q_{kt} =$

X_{kt}/P_t . Let the elasticity of real expenditures be given by

$$\zeta = -\frac{\partial \log Q_{kt}}{\partial \log P_t}.$$

For nominal expenditures $X_{kt} = P_t Q_{kt}$, it follows that $\frac{\partial \log X_{kt}}{\partial \log P_t} = 1 - \zeta$. Thus, in holding X_{kt} fixed, we implicitly impose $\zeta = 1$. When price index P_t rises by 1%, real expenditures Q_{kt} fall by 1%. That is, while nominal expenditures X_{kt} are fixed, real expenditures Q_{kt} respond directly to changes in battery prices.

Furthermore, battery quantities d_{jkt} reflect both changes in real expenditures and the reallocation of expenditures across technologies, which we estimate. Equation 4 governs these quantity responses. When battery prices p_{jt} rise proportionally and $\beta_j = 0$ for all technologies j , homotheticity implies that expenditure shares w_{jkt} remain fixed. Battery quantities d_{jkt} fall across technologies in proportion to the rise in prices. When battery prices p_{jt} rise proportionally but $\beta_j \neq 0$ for some technology j , real expenditures Q_{kt} decline and generate income effects through the X_{kt}/P_t term. Expenditure shares adjust accordingly. When battery prices p_{jt} increase differentially, relative prices change and induce substitution across technologies through the $\log p_{jt}$ terms. Expenditure shares adjust further as demand reallocates across technologies.

Timing. Our baseline specification uses quarterly expenditure shares and contemporaneous quarterly prices. However, battery procurement contracts and technology-switching frictions may delay demand adjustments. We assess sensitivity to our timing choice with two alternative specifications. First, we retain quarterly expenditure shares but measure prices with two-quarter rolling averages. Second, we aggregate expenditure shares and prices to two-quarter intervals. Table B1 compares the implied demand elasticities. All three specifications produce the same qualitative patterns of substitution. The rolling-average specification yields elasticities that are smaller in magnitude, while two-quarter aggregation yields elasticities that are larger in magnitude. Our baseline estimates lie between these two alternatives.

Instruments. We construct our battery price instruments from the prices of aluminum, phosphoric acid, iron, and manganese. We argue that exclusion is plausible because EV batteries account for a negligible fraction of total demand for these minerals. However, aluminum and iron are also important inputs into the production of

Table B1: Demand elasticities for alternative timing specifications

Elasticity	One-quarter frequency	One-quarter frequency, rolling-average prices	Two-quarter frequency
ϵ_{LL}	-6.23 (0.85)	-5.34 (1.77)	-8.61 (2.85)
ϵ_{LN}	4.01 (0.84)	3.14 (1.70)	6.37 (2.92)
ϵ_{NL}	0.87 (0.14)	0.72 (0.29)	1.28 (0.48)
ϵ_{NN}	-1.67 (0.14)	-1.52 (0.28)	-2.07 (0.49)

We present Marshallian demand elasticities for three timing specifications. First, our baseline specification uses quarterly prices and expenditure shares. Second, we use quarterly expenditure shares with two-quarter rolling averages for prices. The price in a given quarter reflects the price in the current quarter and the previous one. Third, we aggregate expenditure shares and prices to two-quarter intervals. We cluster standard errors by quarter, allowing for serial correlation with a lag of one quarter. We compute Marshallian elasticity standard errors using the delta method.

Table B2: Demand elasticities for alternative instrument sets

Elasticity	With Al and Fe	Without Al	Without Fe	Without Al or Fe
ϵ_{LL}	-6.23 (0.85)	-6.55 (1.02)	-6.17 (0.82)	-6.50 (0.99)
ϵ_{LN}	4.01 (0.84)	4.33 (1.03)	3.95 (0.81)	4.28 (0.99)
ϵ_{NL}	0.87 (0.14)	0.93 (0.17)	0.86 (0.14)	0.92 (0.16)
ϵ_{NN}	-1.67 (0.14)	-1.72 (0.17)	-1.66 (0.13)	-1.71 (0.17)

We present Marshallian demand elasticities for four instrument sets. First, our baseline instrument set includes aluminum, phosphoric acid, iron, and manganese. Second, we exclude aluminum. Third, we exclude iron. Fourth, we exclude both aluminum and iron. We cluster standard errors by quarter, allowing for serial correlation with a lag of one quarter. We compute Marshallian elasticity standard errors using the delta method.

vehicle bodies. A potential threat to exclusion is that their prices may affect battery demand through non-battery vehicle costs, beyond their role as battery cost shifters. Table B2 shows that our results are robust to excluding aluminum prices, iron prices, or both from the construction of our instruments.

C Supply

We present key derivations, and we show that our framework accommodates a more general cost structure. We extend the model to capture dynamics from ore degradation, as well as measurement error in our cost data.

C.1 Derivations

Euler equation. Mine i chooses an extraction path $\{x^{it}\}_{t \geq 0}$ to maximize the present value of expected profits, subject to the reserves constraint. The Lagrangian is

$$\mathcal{L}^{it} = \sum_{t'=t}^{\infty} \beta^{t'-t} \mathbb{E}^{it} \left[mr^{it'} x^{it'} - C^{it'} + \lambda^{it'} (R^{it'} - x^{it'} - R^{it'+1}) \right].$$

The first-order conditions with respect to x^{it} and R^{it+1} are

$$\lambda^{it} = mr^{it} - mc^{it}, \quad \lambda^{it} = \beta \mathbb{E}^{it} [\lambda^{it+1}].$$

Substitution gives Euler equation 13.

$$mr^{it} - mc^{it} = \beta \mathbb{E}^{it} [mr^{it+1} - mc^{it+1}]$$

Utilization. Substituting equation 12 into $\lambda^{it} = mr^{it} - mc^{it}$,

$$\lambda^{it} = mr^{it} - \alpha^i - \kappa^m u^{it} - \varepsilon^{it}.$$

We apply equation 11, and we rearrange to obtain equation 14 for utilization.

$$u^{it} = \frac{2}{\kappa^m} (mr^{it} - ac^{it} - \lambda^{it})$$

Myopia and linear costs. We consider the special case with $\beta = 0$ and $\kappa^m = 0$. The Euler equation specializes to $mr^{it} = ac^{it}$, which yields a corner solution. Mines produce at capacity if $mr^{it} > ac^{it}$, and they produce zero otherwise.

C.2 Generalized cost structure

We show that our approach accommodates the more general cost structure

$$C^{it} = F(x^{it}; \theta) + H(x^{it}, \varepsilon^{it})$$

for supply decisions x^{it} , cost parameters θ , and cost shocks ε^{it} . Function F captures the parametric components of costs, and function H captures unobserved shocks. We allow shocks ε^{it} to interact with supply decisions x^{it} . Average and marginal costs are

$$ac^{it} = f(x^{it}; \theta) + h(x^{it}, \varepsilon^{it}), \quad (21)$$

$$mc^{it} = f(x^{it}; \theta) + f'(x^{it}; \theta) x^{it} + h(x^{it}, \varepsilon^{it}) + h'(x^{it}, \varepsilon^{it}) x^{it}, \quad (22)$$

where $f = F/x$, $h = H/x$, and primes denote derivatives with respect to x . Our approach is to invert cost shocks from average cost data and to substitute into the Euler equation, eliminating ε^{it} . Identification requires three conditions.

Separability. Costs separate into components F and H , as written above.

Invertibility. The unobserved component takes the form

$$h(x^{it}, \varepsilon^{it}) = b(\varepsilon^{it}) v(x^{it}).$$

By equation 21, we can invert to express $b(\varepsilon^{it})$ as a function of observed (ac^{it}, x^{it}) .

$$b(\varepsilon^{it}) = \frac{ac^{it} - f(x^{it}; \theta)}{v(x^{it})}$$

Differentiating and substituting,

$$h'(x^{it}, \varepsilon^{it}) = \frac{v'(x^{it})}{v(x^{it})} [ac^{it} - f(x^{it}; \theta)].$$

Equation 22 becomes

$$mc^{it} = f(x^{it}; \theta) + f'(x^{it}; \theta)x^{it} + \underbrace{ac^{it} - f(x^{it}; \theta)}_{h(x^{it}, \varepsilon^{it})} + \underbrace{\frac{v'(x^{it})}{v(x^{it})} h(x^{it}, \varepsilon^{it})}_{h'(x^{it}, \varepsilon^{it})} x^{it}.$$

Collecting terms, we express marginal costs as a function of terms that do not depend on ε^{it} . We thus eliminate ε^{it} from the Euler estimating equation.

$$mc^{it} = ac^{it} + \frac{v'(x^{it})}{v(x^{it})} ac^{it} x^{it} + \left[f'(x^{it}; \theta) - f(x^{it}; \theta) \frac{v'(x^{it})}{v(x^{it})} \right] x^{it}. \quad (23)$$

Nondegeneracy. Only the last term of equation 23 depends on θ . Thus, identification of θ requires

$$f(x^{it}; \theta) \neq a(\theta) v(x^{it}) \quad \forall x^{it}.$$

If this condition fails, we have

$$f'(x^{it}; \theta) = f(x^{it}; \theta) \frac{v'(x^{it})}{v(x^{it})},$$

which eliminates θ from equation 23 and thus from the Euler estimating equation. Intuitively, if f and v are proportional, then variation in x^{it} cannot separately identify $(\theta, \varepsilon^{it})$ because $ac^{it} = [a(\theta) + b(\varepsilon^{it})] v(x^{it})$.

Our baseline specification satisfies all three conditions. Suppressing superscripts, the main text has parameters $\theta = \{\alpha, \kappa\}$ and functions

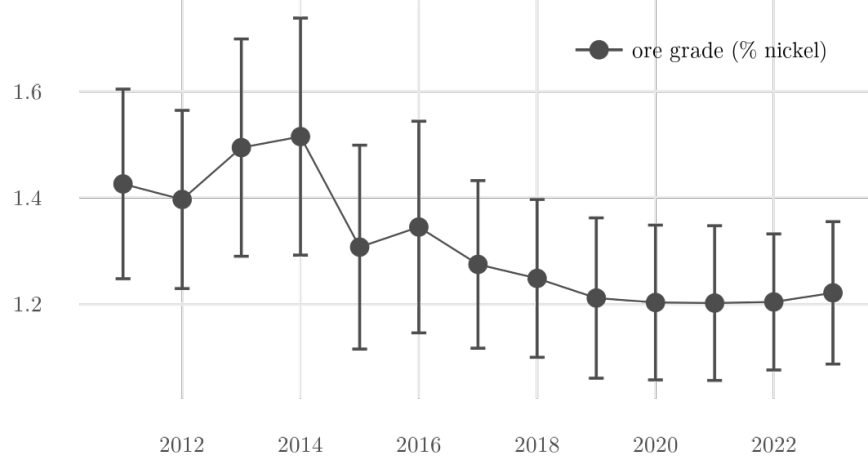
$$F(x; \theta) = \left(\alpha + \frac{\kappa}{2\bar{x}} x \right) x, \quad H(x, \varepsilon) = \varepsilon x.$$

Separability holds with $f(x; \theta) = \alpha + \frac{\kappa}{2\bar{x}} x$ and $h(x, \varepsilon) = \varepsilon$. Invertibility holds because $b(\varepsilon) = \varepsilon$ and $v(x) = 1$ yield $\varepsilon = ac - f(x; \theta)$. Nondegeneracy holds because f depends on x , while v does not.

C.3 Ore degradation

We can allow ore quality to degrade with extraction. In the spirit of [Aguirre-gabiria and Luengo \(2016\)](#), we specify future grade as a function of current grade and current production. Current grade acts as a state variable that summarizes cumula-

Figure C1: Nickel ore grade over time



We plot world average nickel ore grades with 95% confidence intervals. We average across mines, weighting by output.

tive extraction.

$$\log g^{it+1} = \log g^{it} - \delta \log s^{it} + \varphi^i + \psi^t + \epsilon^{it}, \quad (24)$$

where δ is the elasticity of degradation, φ^i and ψ^t are mine and year fixed effects, and ϵ^{it} is the error term. We estimate this specification separately for each mineral, and we obtain $\hat{\delta}$ estimates of 0.02 for lithium, 0.06 for nickel, and 0.09 for cobalt. Each is small in magnitude. Figure C1 illustrates ore grade decay over time for nickel, again revealing limited degradation in the data. Our baseline model therefore imposes constant ore grade g^i .

It is conceptually straightforward to accommodate ore degradation, by which current extraction reduces future marginal revenues. We simply augment the Euler equation to capture this additional dynamic cost.

$$mr^{it} - mc^{it} - \beta \mathbb{E}^{it}[mc_d^{it+1}] = \beta \mathbb{E}^{it}[mr^{it+1} - mc^{it+1} - \beta mc_d^{it+2}],$$

where mc_d^{it+1} is the marginal degradation cost. It captures the impact of current extraction on future revenue through the deterioration of ore grade.

To construct mc_d^{it+1} , we substitute production $s^{it} = g^{it}x^{it}$ into grade evolution equation 24, and we differentiate to obtain the elasticity of future grade g^{it+1} with

respect to current extraction x^{it} .

$$\frac{\partial \log g^{it+1}}{\partial \log x^{it}} = -\delta < 0, \quad \frac{\partial g^{it+1}}{\partial x^{it}} = -\delta \frac{g^{it+1}}{x^{it}} < 0$$

The cost of degradation is the resulting reduction of future extraction revenue. Since revenue $p^{mt+1}g^{it+1}x^{it+1}$ is linear in grade, each unit of grade lost reduces revenue by $p^{mt+1}x^{it+1}$.

$$mc_d^{it+1} = \delta \frac{g^{it+1}}{x^{it}} p^{mt+1} x^{it+1} > 0.$$

We use this expression to construct mc_d^{it+1} from our estimated $\hat{\delta}$ and our data on grades g^{it} , prices p^{mt} , and extraction x^{it} . Estimating equation 17 becomes

$$\Delta p^{mt} g^i - \Delta a c^{it} - \beta \Delta m c_d^{it+1} = \frac{1}{2} \kappa^m \Delta u^{it} + \Delta \tau^{km} g^i + \Delta \tau^{om} g^i + \eta^{it}.$$

C.4 Measurement error

Suppose we observe average costs with classical measurement error ν^{it} , such that $\hat{a}c^{it} = ac^{it} + \nu^{it}$ with $\mathbb{E}[\nu^{it}] = 0$. Estimating equations 17 and 18 become

$$\begin{aligned} \Delta p^{mt} g^i - \Delta \hat{a}c^{it} &= \frac{1}{2} \kappa^m \Delta u^{it} + \Delta \tau^{km} g^i + \Delta \tau^{om} g^i + \eta^{it} - \Delta \nu^{it}, \\ \hat{a}c^{it} - \frac{1}{2} \hat{\kappa}^m u^{it} &= \alpha^i + \varepsilon^{it} + \nu^{it}. \end{aligned}$$

Measurement error adds to the error terms but does not bias our estimates, as the composite error terms $(\eta^{it} - \Delta \nu^{it})$ and $(\varepsilon^{it} + \nu^{it})$ remain uncorrelated with our regressors. Intuitively, it is straightforward to accommodate mismeasurement of average costs because they enter our estimating equations on the left-hand sides, and not on the right-hand sides. Without measurement error, we recover expectational error η^{it} as residuals in equation 17, and we recover cost shocks ε^{it} as residuals in equation 18. With measurement error, the residuals instead recover the composite error terms.

D Counterfactuals

We describe how we solve the model for our counterfactual analysis, present robustness checks, and report additional results.

D.1 Calibration and assumptions

Projected battery expenditures. We construct a path for regional battery expenditures X_{kt} beyond the estimation sample, using forecasts of battery quantities by cathode chemistry from BMI (2025) and observed 2024 battery prices. We proceed in three steps.

First, we begin from observed 2024 Q1 battery expenditures in each region. Let $q_{jk,2024Q1}$ denote observed battery demand in MWh for battery j and region k in 2024 Q1, and let $p_{j,2024}$ denote the corresponding 2024 battery price. We aggregate across batteries to obtain base-year regional expenditure levels.

$$X_{k,2024Q1} = \sum_j p_{j,2024} q_{jk,2024Q1}$$

Second, we use BMI forecasts of global battery demand by cathode chemistry. BMI reports forecasted future quantities in MWh at the chemistry-year level. We map these chemistries into our battery technologies, and we aggregate to obtain projected global quantities Q_{jt} for each battery j and year t . We distribute these projected global quantities across regions using 2024 Q1 regional shares for each battery: $\omega_{jk} = \frac{q_{jk,2024Q1}}{\sum_{k'} q_{jk',2024Q1}}$. Given these shares, the projected regional annual quantities are

$$\hat{q}_{jkt} = \omega_{jk} Q_{jt}.$$

Third, we convert projected quantities into annual expenditures using 2024 prices. The expenditure path is driven by projected growth in battery quantities, holding fixed both regional composition within batteries and the battery prices used to value these quantities.

$$\hat{X}_{kt} = \sum_j p_{j,2024} \hat{q}_{jkt}$$

Non-EV mineral demand. Our counterfactuals require a path of non-EV mineral demand ν^{mt} for each mineral m . We impose a linear demand function, calibrated from 2024 data and held fixed over time.

$$\nu^{mt}(p^{mt}) = a^m + b^m p^{mt}$$

We recover 2024 non-EV demand $\bar{\nu}^m$ as the residual between observed 2024 mineral output and observed 2024 EV battery demand, such that $\bar{\nu}^m = s^{m,2024} - \sum_{j \in \mathcal{J}} r_j^m d_{j,2024}$. We calibrate the slope so that the elasticity of non-EV demand at the 2024 price and quantity is -0.1 , which is consistent with recent estimates of mineral demand elasticities (Shojaeddini et al. 2025).

$$b^m = -\frac{0.1 \bar{\nu}^m}{p^{m,2024}}, \quad a^m = 1.1 \bar{\nu}^m$$

Mine-level reserves. While we do not require reserves data for the purpose of estimating mine costs in Section 5, we do require these reserves at the mine level for computing counterfactuals in Section 7. Our mining data have incomplete coverage of reserves, and so we allocate world reserves across mines to impute missing mine-level reserves. We measure world reserves with data from USGS. For each mineral, we allocate world reserves across mines in proportion to the maximum production that we observe in our sample for each mine. USGS world reserves are measured in units of mineral content, rather than crude ore extracted as in Section 5, and so we divide by ore grade g^i to convert. Reserves $R^{i,2024}$ for each mine i are

$$R^{i,2024} = \frac{\bar{s}^i}{\sum_{i' \in \mathcal{I}^{m(i)}} \bar{s}^{i'}} \frac{\bar{R}^{m(i),2024}}{g^i},$$

where $\bar{R}^{m,2024}$ is world reserves of mineral m in 2024, and $\bar{s}^i$ is the maximum observed production of mine i in our sample.

To accommodate reserve growth, we assume world reserves for a mineral grow at a constant annual rate.

$$\bar{R}^{m,t+1} / \bar{R}^{mt} = \rho^m$$

We calibrate the growth rate ρ^m to be the average observed in our reserves data, provided in Table D1. Our mine data provide information on mines under development,

including mine location, predicted production, and predicted entry date. We allocate reserve growth in year t , $\bar{R}^{mt} - \bar{R}^{m,t-1}$, to mines that enter in year t in proportion to predicted production. If no new mines enter in year t for a given mineral, we increase all mines' reserves for that mineral proportionally by the calibrated growth rate $\hat{\rho}^m$.

Table D1: Historical reserves growth

Mineral	Growth
Lithium	8.3%
Nickel	4.9%
Cobalt	4.3%

Growth rates are determined by taking the growth in reserves measured by USGS for each mineral from 2015 until the end of our sample and computing the implied average annual growth rate.

Recovered parameters. Table D2 reports recovered midstream wedges and non-EV mineral demand.

Table D2: Recovered parameters

(a) Midstream wedges		(b) Non-EV mineral demand		
Battery	μ_j	Mineral	$\nu^m = a^m + b^m p^m$	
			a^m	b^m
L	46,155	Lithium	1,024,736	−23.0
N	46,679	Nickel	4,956,620	−106
		Cobalt	221,541	−5.97

We report μ_j in USD/MWh, a^m in tonnes, and b^m in tonnes per USD/tonne.

D.2 Robustness

Isolating joint use. Figure 9 simulates Australian lithium and Indonesian nickel policies with and without joint use. We shut down joint use by setting recipe term $r_N^\ell = 0$, which eliminates lithium use in battery N . In this case, nickel is used exclusively in battery N , and lithium is used exclusively in battery L . A potential concern with this approach is that it confounds the effects of joint use with two mechanical price effects. First, setting $r_N^\ell = 0$ lowers the battery N price by removing

Table D3: Joint use with adjusted wedges

Policy	Adoption	With joint use	Without joint use	
		Recovered wedges (%)	Fixed wedges (%)	Adjusted wedges (%)
Australian lithium	Battery L	−3.14	−3.56	−3.44
	Battery N	−1.51	1.45	1.40
	Total	−2.30	−0.91	−0.91
Indonesian nickel	Battery L	0.52	0.48	0.44
	Battery N	−1.21	−1.34	−1.22
	Total	−0.36	−0.48	−0.41

We simulate unilateral lithium and nickel policies with and without joint use. Joint use is given by our baseline battery recipes with $r_N^\ell > 0$. We recover wedges (μ, ν) from the data and hold them fixed in our counterfactual, where μ captures midstream costs and potential markups and ν captures non-EV demand. We shut down joint use by setting $r_N^\ell = 0$ in our battery recipes. We do so holding wedges (μ, ν) at their recovered values, and again adjusting these wedges to match baseline mineral prices. All changes are relative to the baseline equilibrium with the same recipes and wedges, but no restrictive policies. Battery L is LFP, and battery N is NMC and NCA.

a costly input to production. Second, setting $r_N^\ell = 0$ directly lowers the lithium price by removing a source of lithium demand.

We show that our results are robust to these concerns. We do so with two further adjustments that eliminate the mechanical price effects. In particular, we adjust midstream wedges μ_{jt} and non-EV demand ν^{mt} such that baseline mineral prices still clear markets for each mineral in each period. The adjusted values $(\hat{\mu}_{jt}, \hat{\nu}^{mt})$ satisfy

$$s^{mt}(\bar{p}^{mt}, \bar{p}^{mt+1}, \dots, \bar{p}^{mT}) = \sum_{j \in \mathcal{J}} \hat{r}_j^m d_{jt}(\hat{\mathbf{p}}_t) + \hat{\nu}^{mt}, \quad \hat{p}_{jt} = \sum_{m \in \mathcal{M}} \hat{r}_j^m \bar{p}^{mt} + \hat{\mu}_{jt}$$

for baseline mineral prices $\bar{p}^{mt}$ and counterfactual recipes $\hat{r}_j^m$. These adjustments restore the price of battery N and the demand for lithium to their baseline values when $r_N^\ell > 0$. Table D3 shows that these adjustments have little effect on our results.

Terminal year. Table D4 reports 2024 prices and quantities for each mineral under the baseline counterfactual, which we re-solve using different terminal years. We obtain similar results in each case.

Table D4: Baseline equilibrium by terminal year

Mineral	Outcome	End 2035	End 2040	End 2045
Lithium	Price	4,607	5,940	7,159
Lithium	Quantity	1,424,782	1,387,051	1,352,299
Nickel	Price	4,498	4,562	4,609
Nickel	Quantity	4,881,358	4,869,912	4,860,085
Cobalt	Price	4,449	4,512	5,057
Cobalt	Quantity	291,898	290,391	285,989

Prices are in USD per tonne, and quantities are in tonnes. Each column solves for the baseline equilibrium with the indicated terminal year. Rows report 2024 prices and quantities for each mineral.

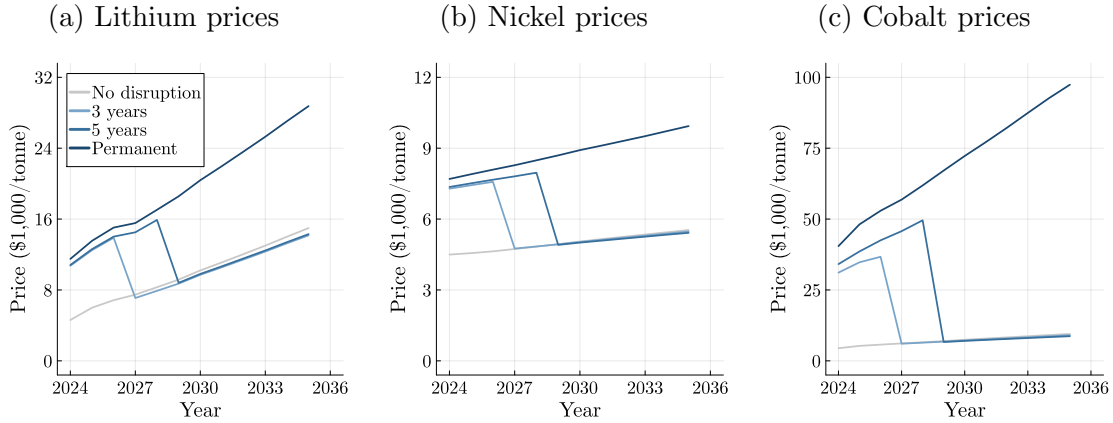
D.3 Additional results

Temporary supply disruptions. Table 2 simulates permanent supply disruptions to mineral production. Figure D1 evaluates more short-lived disruptions, plotting the price effects over time of removing each mineral’s top producer. Temporary disruptions generate smaller price increases than permanent disruptions. When disruptions are temporary, the largest producer eventually resumes production, and this future production drives down future prices. Lower future prices reduce the shadow cost of reserves for undisrupted producers, who respond by increasing current production. Higher production dampens price increases during the disruption. By contrast, permanent disruptions also reduce future production, raising future prices and thus the shadow cost of reserves. Prices spike strongly as undisrupted producers hold reserves for future extraction.

Foreign ownership. Foreign ownership may influence the set of profits that enter a mineral-producing country’s objective function, as well as the set of mines subject to its policy. Table D5 compares Australian lithium policy under three treatments of foreign ownership. First, when Australia taxes all lithium mines in Australia and values the profits of all such mines in its objective function, the optimal tax is \$6,322 per tonne. Second, when Australia continues to tax all mines in Australia but values only the profits of domestically owned mines, the optimal tax rises to \$8,740 per tonne. Because the government still collects revenue from foreign-owned mines but no longer places weight on their profits, it has a stronger incentive to tax them. Third,

when Australia both taxes and values profits only for domestically owned mines, the optimal tax falls to \$2,671 per tonne. In this case, the tax base is smaller, and every taxed mine’s profits enter the government’s objective. The resulting optimal tax is the lowest among the three scenarios.

Figure D1: Supply disruptions over time



For each mineral, we remove the top producer for a particular number of years and compute the resulting mineral prices in each year. We simulate no disruption, a three-year disruption, a five-year disruption, and a permanent disruption. Table 2 simulates permanent disruptions. We assume firms have full knowledge of the length of the disruption.

Table D5: Foreign ownership and Australian lithium policy

Mines taxed by Australian policy	Profits included in Australian objective	Tax (\$/t)	Lithium price (%)	Lithium quantity (%)	Battery <i>L</i> quantity (%)	Battery <i>N</i> quantity (%)
All Australian	All Australian	6,322	29.0	−3.8	−3.1	−1.5
All Australian	Domestically owned	8,740	42.5	−5.6	−4.6	−2.2
Domestically owned	Domestically owned	2,671	6.8	−0.9	−0.8	−0.4

Price and quantity columns report average percentage changes relative to baseline over the counterfactual horizon. Tax revenue from all mines subject to the tax enters the Australian welfare objective. “All Australian” refers to all lithium mines located in Australia, regardless of ownership. Battery *L* is LFP, and battery *N* is NMC and NCA.

E Notation

Symbol	Description
i	indexes mines
j	indexes battery technologies
k	indexes regions/countries
m	indexes minerals
t	indexes time
o	indexes mine-owner countries
$\mathcal{I}^m$	set of mines producing mineral m
$\mathcal{J}$	set of battery technologies
$\mathcal{K}$	set of regions
$\mathcal{M}$	set of minerals
p_{jt}	price per MWh of battery j in period t
P_t	battery price index in period t
X_{kt}	battery expenditure of EV manufacturers in region k and period t
w_{jkt}	expenditure share of battery j in region k and period t
d_{jkt}	demand for battery j in region k and period t
α_{jt}	battery technology quality in demand
β_j	battery-specific expenditure coefficient in demand
$\gamma_{jj'}$	battery own- and cross-price semi-elasticity in demand
ε_{jkt}	battery demand shock
r_j^m	tonnes of mineral m per MWh of battery j
μ_{jt}	midstream wedge for battery j
p^{mt}	price per tonne of mineral m in period t
d^{mt}	world demand for mineral m in period t
s^{mt}	world supply of mineral m in period t
ν^{mt}	non-EV demand for mineral m in period t
x^{it}	ore extraction by mine i in period t
s^{it}	mineral output of mine i in period t
g^i	ore grade of mine i
τ^i	price wedge faced by mine i
$\bar{x}^i$	extraction capacity of mine i
u^{it}	utilization rate of mine i in period t
R^{it}	remaining reserves of mine i in period t
C^{it}	total cost of mine i in period t
α^i	intercept of marginal cost for mine i
κ^m	slope of marginal cost for mineral m
ε^{it}	mine-level cost shock
λ^{it}	shadow cost of reserves for mine i in period t
β	discount factor